Magic Gems: Magic Squares as Orthogonality Conditions on a Height Function

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Abstract

We study *Magic Gems*: the convex hull of the points obtained by placing the entries of an $n \times n$ magic square as heights over a centred grid. The representation is D_4-invariant, so each essentially different square has one gem. Reading the centred entries as a height function z, the magic conditions are exactly the orthogonality of z to a $2n$-dimensional space of centred line indicators; we treat this as a change of viewpoint, not a theorem, and use it to obtain: (i) a moment hierarchy. For real height functions, $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$ for $m \le k$ is equivalent to equal row and column sums exactly when $k \ge n-1$ (a Vandermonde fact). For permutations of $1, \dots, n^2$, integrality lowers the threshold to at most $n-2$ for every $n \ge 14$. Loly's moment-of-inertia invariance is a consequence of the degree-two level of the hierarchy, and the moment of inertia of a uniformly random arrangement has mean equal to Loly's invariant value, with an explicit variance; (ii) exact mean and variance of two energies on a uniformly random arrangement, with a limit law by Hoeffding's combinatorial central limit theorem; (iii) an exact single-swap formula, giving strict local minimality and gaps d^2m/n^4; (iv) the exact count 14,639,440 of 4×4 arrangements satisfying the four low-mode conditions; (v) an exact computation of the geometry of the 880 order-four gems (39 f-vectors, 352 combinatorial types, 616 congruence classes, which are exactly the orbits of the complement map $S \mapsto 17 - S$) and a computer-assisted proof that Siamese gems have twelve vertices.

Keywords: magic squares, convex polyhedra, permutation statistics, moment of inertia, combinatorial geometry

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu appears in Chinese sources more than two millennia old, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and systematic constructions for every order have been known since the seventeenth century [2, 18]. The modern literature treats them from several directions: enumeration and the geometry of the associated polytopes [21, 8, 1, 4, 3], spectral and matrix properties [15, 16, 17, 5], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [13, 19, 20].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \dots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $M(n) = n(n^2+1)/2$, and *semi-magic* if only the rows and columns do.

This paper studies a three-dimensional representation of an arrangement: place the centred entry $s_{ij} - (n^2+1)/2$ as a height over the centred cell $(j - \frac{n-1}{2},\ \frac{n-1}{2} - i)$ and take the convex hull of the n^2 points, the *Magic Gem* of S (Section 3). The representation is invariant under the dihedral

group D_4 of the square, so each essentially different magic square has a single gem (Proposition 3.3); an interactive web application displays the 880 essentially different order-four squares and renders the gem of a selected one (points, convex hull, vectors from the centre, coordinate axes; rotatable, with orthographic views) at `https://kylemath.github.io/MagicGemWebpage/`. The convex hull itself is the object of Sections 5.5–5.6 (the geometry of the 880 order-four gems and of the Siamese gems); everywhere else the object is the height function z. Once the entries are read as a function on the grid, every line-sum condition "row k sums to $M(n)$" becomes "z is orthogonal to the centred indicator of row k", i.e. $\mathrm{Cov}(R_k, Z) = 0$ for the empirical covariance over the n^2 cells. A magic square is therefore an arrangement whose height function is orthogonal to a fixed $2n$-dimensional space W of test functions, and the natural *energy* $E_{\mathrm{full}}(S) = \sum_L \mathrm{Cov}(\mathbf{1}_L, Z)^2$ over the $2n+2$ lines vanishes exactly on magic squares (Lemma 4.7). We are explicit that this is a reformulation of the definition and not a theorem about magic squares (Remark 4.8); its value is that it suggests questions whose answers are not visible in the line-sum formulation, and the paper is organised around those answers.

Contributions.

1. *The moment hierarchy* (Proposition 4.10, Theorem 4.14). Replacing the line indicators by the polynomial test functions x^m, y^m gives, for each degree k, the conditions $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$ for $m \le k$. For real-valued height functions these are equivalent to equal row and column sums if and only if $k \ge n-1$; this is the invertibility of a Vandermonde matrix and we state it as a proposition. The result of the section is Theorem 4.14: for permutations of $1, \dots, n^2$, integrality and the bound on line sums force the threshold $k^*(n)$ down to at most $n-2$ for every $n \ge 14$, whereas $k^*(n) = n-1$ for $3 \le n \le 13$ (computer-verified witnesses, Remark 4.15). The "low-mode" energy E_{low} of earlier versions of this paper is the degree-one truncation. Loly's invariance of the moment of inertia [13], which Loly derived from the row and column sums alone and so stated for semi-magic squares, is the single condition $\mathrm{Cov}(Z, X^2 + Y^2) = 0$: implied by the degree-two conditions, but much weaker than them (Corollary 4.12). The same identity gives the null distribution of the moment of inertia of a random arrangement (Corollary 4.13).
2. *Exact statistics of a random arrangement* (Proposition 5.1). $\mathbb{E}[E_{\mathrm{low}}] = (n^2+1)(n^4 - n^2 + 12n - 12)/(72n^2)$ and $\mathbb{E}[E_{\mathrm{full}}] = (n^4-1)/(6n^2)$, with exact variances; $E_{\mathrm{low}}/\mathbb{E}[E_{\mathrm{low}}] \to \mathrm{Exp}(1)$ by Hoeffding's combinatorial central limit theorem. Monte Carlo with 10^8 samples per order agrees for $n \le 8$ to within 5×10^{-4} relative error.
3. *The single-swap formula* (Theorem 4.17). Exchanging two entries of a magic square that differ by d raises E_{full} by exactly $d^2 m/n^4$, where $m \in \{2, \dots, 6\}$ counts the lines separating the two cells; every magic square is a strict local minimum, with minimum gap between $2/n^4$ and $6/n^4$.
4. *The low-mode class for $n = 4$* (Proposition 5.4). Exactly 14,639,440 of the 16! arrangements satisfy $E_{\mathrm{low}} = 0$, 7,040 of them magic; the count is a meet-in-the-middle enumeration and agrees with a Gaussian local-limit estimate to within 20%.
5. *Geometry of the 880 order-four gems* (Section 5.5). 8 to 16 hull vertices, 39 f-vectors, 352 combinatorial types and 616 congruence classes, which are exactly the orbits of the complement map $S \mapsto 17 - S$; no two gems have the same hull-vertex set. The Siamese squares of every odd order $n \ge 5$ have gems with exactly twelve vertices (Proposition 5.7, computer-assisted proof).

What is and is not new. The polyhedral geometry of magic squares is classical: nonnegative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope and their number is a polynomial in t [21, 8]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [4, 3]; the cones of magic squares

and cubes have Hilbert-basis descriptions [1]. The Magic Gem is a different object: not the polytope whose lattice points are the squares, but a polyhedron attached to one square. The orthogonality reading of the constraints is elementary and implicit in Loly's work, and the hierarchy itself is elementary: that the axial moments $\sum_{ij} s_{ij} x_j^m$ of a semi-magic object vanish for every m is, in physical language, the vanishing of its multipole moments, as Rogers and Loly observed for the dipole and quadrupole moments of a magic cube [20]. What is new is the converse threshold and its permutation refinement (Theorem 4.14), the exact random-arrangement moments, the swap formula, the exact low-mode count, and the exact geometry of the 880 gems. We make no claim that the energy reformulation by itself is a result.

Corrections relative to version 4 (arXiv:2512.09170v4).

(a) The complete energy used there omitted one row and one column and was not D_4-invariant; all statistics now use the symmetric sum (6).

(b) The sampled statistics (means 4.44, 12.04, 26.00), the "landscape entropy" and the quadratic fits had the diagonal terms weighted by n^2; withdrawn, replaced by Proposition 5.1.

(c) The "trimodal" swap gaps and the "$1/n^2$" minimum-gap scaling were artifacts: gaps are $d^2 m/n^4$ and $\Delta_{\min} = \Theta(n^{-4})$ with constants 2, 6 (Corollary 4.19).

(d) The 24 local minima of E_{full} for $n = 3$ referred to the asymmetric energy; the symmetric one has 16 (Section 5.1).

(e) "$E_{\text{low}} = 0$ characterises magic (or semi-magic) squares" is withdrawn (Remark 4.3, Proposition 5.4); $P(E_{\text{low}} = 0)$ for $n = 4$ is 7.0×10^{-7}, not "below 10^{-8}", and the statement that 6×10^7 samples for $n = 4$ and 4×10^8 for $n = 5$ contained no zero is unsupported by the archived data, which hold 10^5 samples per order; the cause cannot be reconstructed from the repository (Section 5.4).

(f) Table 2 of v4 ("typical" 16 hull vertices for $n = 5$, volume 95.0 for $n = 4$) is replaced by exact values for named squares (Section 5.5); the hardware statement is corrected (Section A).

Section 2 collects the facts we use; Section 3 defines the gem; Section 4 contains the projection lemma, the hierarchy and the swap formula; Section 5 the computations; Section 6 what the viewpoint does and does not buy, local search, and open questions; the appendices the methods, the variance closed forms, the witnesses and the remaining steps of the Siamese proof.

2 Background

2.1 Enumeration and symmetry

There are 8 magic squares of order 3, all images of the Lo Shu under the dihedral group D_4 of the square (four rotations, four reflections); there are 7,040 of order 4, forming 880 D_4-orbits first listed by Frénicle de Bessy (published posthumously in 1693) and verified many times since [3]; and 275,305,224 of order 5, counted by Schroeppel in 1973 and reported by Gardner [9], i.e. 34,413,153 orbits. For $n \geq 6$ only statistical estimates are known; see Trump's compilation [22]. Since D_4 maps lines to lines and preserves the entry set, it maps magic squares to magic squares, and two squares in the same orbit are called *essentially different* from squares in other orbits. The 880 order-four squares were further sorted by Dudeney into twelve groups according to the pattern of complementary pairs $(a, 17 - a)$ [7]; we use these groups only as a classical reference partition in Section 5.5. Classical constructions (Siamese for odd n, diagonal-complement for $n \equiv 0 \pmod 4$, LUX or Strachey for $n \equiv 2 \pmod 4$) show that magic squares exist for all $n \geq 3$ [2]; we use the Siamese squares as a family on which hull computations can be pushed to large n.

2.2 Covariance and the inertia tensor

For a finite point set $\mathbf{p}_1, \dots, \mathbf{p}_N \in \mathbb{R}^3$ with mean $\bar{\mathbf{p}}$, the covariance matrix is $\Sigma_{jk} = \frac{1}{N}\sum_i (p_{i,j} - \bar{p}_j)(p_{i,k} - \bar{p}_k)$. With unit masses at the points, the inertia tensor about the centre of mass is

$$I_{jk} = \sum_i \big(\|\mathbf{p}_i - \bar{\mathbf{p}}\|^2 \delta_{jk} - (p_{i,j} - \bar{p}_j)(p_{i,k} - \bar{p}_k) \big) = N\big(\delta_{jk} \operatorname{tr}\Sigma - \Sigma_{jk} \big), \tag{1}$$

where δ_{jk} is the Kronecker delta [10]. Thus the off-diagonal entries of I are $-N$ times the covariances and the diagonal entries are N times the sum of the two other variances. Loly's planar moment of inertia [13] is a different object: it places the *value* s_{ij} as a mass at the planar point (x_{ij}, y_{ij}) and computes $\sum s_{ij}(x_{ij}^2 + y_{ij}^2)$. The two are nevertheless related through the gem's point set: Loly's quantity is an affine function of the single degree-two cross-moment $\operatorname{Cov}(Z, X^2 + Y^2)$ of that point set (Corollary 4.12), and this is the only ingredient of the three-dimensional covariance Σ that it uses.

2.3 Two tools from permutation statistics

Let π be a uniformly random permutation of $\{1, \dots, N\}$ and $a, b \in \mathbb{R}^N$ fixed vectors. The quantity $T_a = \sum_i a_i b_{\pi(i)}$ is a *linear permutation statistic*. Its mean is $N\bar{a}\bar{b}$ and, for two such statistics with the same b (Wald–Wolfowitz [23], polarised),

$$\operatorname{Cov}(T_a, T_{a'}) = \frac{1}{N-1} \sum_i (a_i - \bar{a})(a'_i - \bar{a}') \sum_j (b_j - \bar{b})^2. \tag{2}$$

Hoeffding's combinatorial central limit theorem [11, Thm. 3] gives asymptotic normality of $\sum_i c_N(i, \pi(i))$ when, for every integer $r \geq 3$, $N^{-1}\sum_{i,j} |d_N(i,j)|^r$ is $O\big((N^{-1}\sum_{i,j} d_N(i,j)^2)^{r/2}\big)$, d_N being the doubly centred array; for product arrays $c_N(i,j) = a_i b_j$ the condition reduces to a bound on the standardised absolute moments of a and b. In Section 5.2 the energies of a random arrangement are quadratic forms in such statistics; (2) gives their means exactly, and the distribution of a quadratic form in Gaussian variables is computed by Imhof's inversion formula [12].

2.4 Polytopes of magic squares

The polytope literature on magic squares concerns a different polytope from the one studied here. An $n \times n$ nonnegative integer matrix with all row and column sums equal to t is a lattice point of the t-th dilate of the Birkhoff polytope of doubly stochastic matrices, and their number is a polynomial in t of degree $(n-1)^2$ [21, 8, 24]; adding the diagonal conditions and distinct entries leads to the inside-out polytopes of Beck and Zaslavsky [4], which gave the first enumeration of 4×4 magic squares of arbitrary magic sum [3]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; and contingency tables with fixed margins are the same lattice points studied as a sample space [6]. In all of these the magic squares are the *points* of a polytope, whereas the Magic Gem attaches a polyhedron to a *single* square; the two constructions share nothing beyond the word. Spectral and matrix properties of a magic square [15, 16, 17], the singular-value signatures and entropies of Cameron, Rogers and Loly [5], and Loly's physical invariants—the moment of inertia [13], the inertia tensor and the multipole moments of a magic cube [19, 20], and the study of Franklin squares [14]—are closer to our viewpoint: they, too, attach a quantity to one square and ask what it reveals.

3 The Magic Gem

Let $S = (s_{ij})$ be an arrangement of order n, with cells indexed by $i, j \in \{0, \dots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$.

Definition 3.1 (Gem coordinates)**.** To the cell (i, j) assign the point $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ with

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}. \tag{3}$$

The *Magic Gem* of S is the convex hull $\mathcal{G}(S) = \operatorname{conv}\{\mathbf{p}_{ij}\}$.

The horizontal coordinates centre the grid at the origin, with the x-axis along the columns and the y-axis along the rows (top row at positive y); the vertical coordinate centres the entries so that the mean entry $(N+1)/2$ sits at height zero. All three coordinates then have mean zero over the N cells (Lemma 4.1), so the origin is the centroid of the point set. Each $\mathbf{p}_{ij}$ may equally be read as a vector from the centroid, and with unit masses at the points the point set is a rigid body whose centre of mass is the origin; Remark 4.21 records its inertia tensor. Throughout the paper, the function $z = (z_{ij})$ on the grid is called the *height function* of S.

Example 3.2 (Lo Shu)**.** For the Lo Shu square $S = \begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$, $n = 3$ and the centre value is 5, so the nine points are

$$\begin{array}{lll} \mathbf{p}_{00} = (-1, 1, -3), & \mathbf{p}_{01} = (0, 1, 2), & \mathbf{p}_{02} = (1, 1, 1), \\ \mathbf{p}_{10} = (-1, 0, 4), & \mathbf{p}_{11} = (0, 0, 0), & \mathbf{p}_{12} = (1, 0, -4), \\ \mathbf{p}_{20} = (-1, -1, -1), & \mathbf{p}_{21} = (0, -1, -2), & \mathbf{p}_{22} = (1, -1, 3). \end{array}$$

The centre cell maps to the origin and is the only point not on the hull; the gem has f-vector $(8, 16, 10)$ and volume 18 (Figure 1).

Proposition 3.3 (D_4-invariance)**.** *Let $g \in D_4$ act on the grid and let gS be the arrangement with $(gS)_c = s_{g^{-1}c}$. Then $\mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S)$, where $\hat{g}(x, y, z) = (g(x, y), z)$ is the isometry of $\mathbb{R}^3$ that applies the orthogonal map g of the (x, y)-plane and fixes z. In particular $\mathcal{G}(gS)$ and $\mathcal{G}(S)$ are congruent, and the gem is an invariant of the D_4-orbit of S.*

Proof. In the centred coordinates (3), each $g \in D_4$ acts on the cell positions (x_{ij}, y_{ij}) as an orthogonal map of the plane (a rotation by a multiple of 90° or a reflection in a coordinate axis or a diagonal), and $(x_{gc}, y_{gc}) = g(x_c, y_c)$. The point of gS over the cell gc has height $(gS)_{gc} = s_c = z_c + \frac{N+1}{2}$, i.e. the same height as the point of S over c. Hence the point set of gS is $\{(g(x_c, y_c), z_c)\}_c = \hat{g}\{\mathbf{p}_c\}_c$. An affine bijection maps the convex hull of a set to the convex hull of the image, so $\mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S)$; $\hat{g}$ is an isometry because g is orthogonal. □

Since the eight images are congruent, every essentially different magic square has one gem up to congruence; Section 5.5 shows that for $n = 4$ the converse also holds except for the complement map $S \mapsto 17 - S$, which reflects the gem in the plane $z = 0$.

4 Orthogonality conditions on the height function

Throughout, $n \geq 3$, $N = n^2$, and a cell of the $n \times n$ grid is a pair $c = (i, j)$ with $0 \leq i, j \leq n-1$; an arrangement is a bijection $S = (s_{ij})$ from the cells onto $\{1, \ldots, N\}$ (Definition 1.1). We use the centred coordinates of Definition 3.1,

$$x_{ij} = j - \tfrac{n-1}{2}, \qquad y_{ij} = \tfrac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \tfrac{N+1}{2},$$

and refer to $z = z(S)$ as the *height function* of S. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n - 1\}$; there are $2n + 2$ lines, each containing exactly n cells. With $M(n) = n(N+1)/2$ the magic constant, the *deviation* of a line L is

$$\delta_L(S) \;=\; \sum_{c \in L} s_c - M(n) \;=\; \sum_{c \in L} z_c, \tag{4}$$

the second equality because $|L| = n$ and $M(n) = n \cdot \frac{N+1}{2}$. S is *semi-magic* if $\delta_L = 0$ for the $2n$ rows and columns, and *magic* if $\delta_L = 0$ for all $2n + 2$ lines. We write $\rho_k = \delta_{\text{row } k}$, $\gamma_\ell = \delta_{\text{col } \ell}$, δ_{m} and δ_{a} for the row, column, main-diagonal and anti-diagonal deviations.

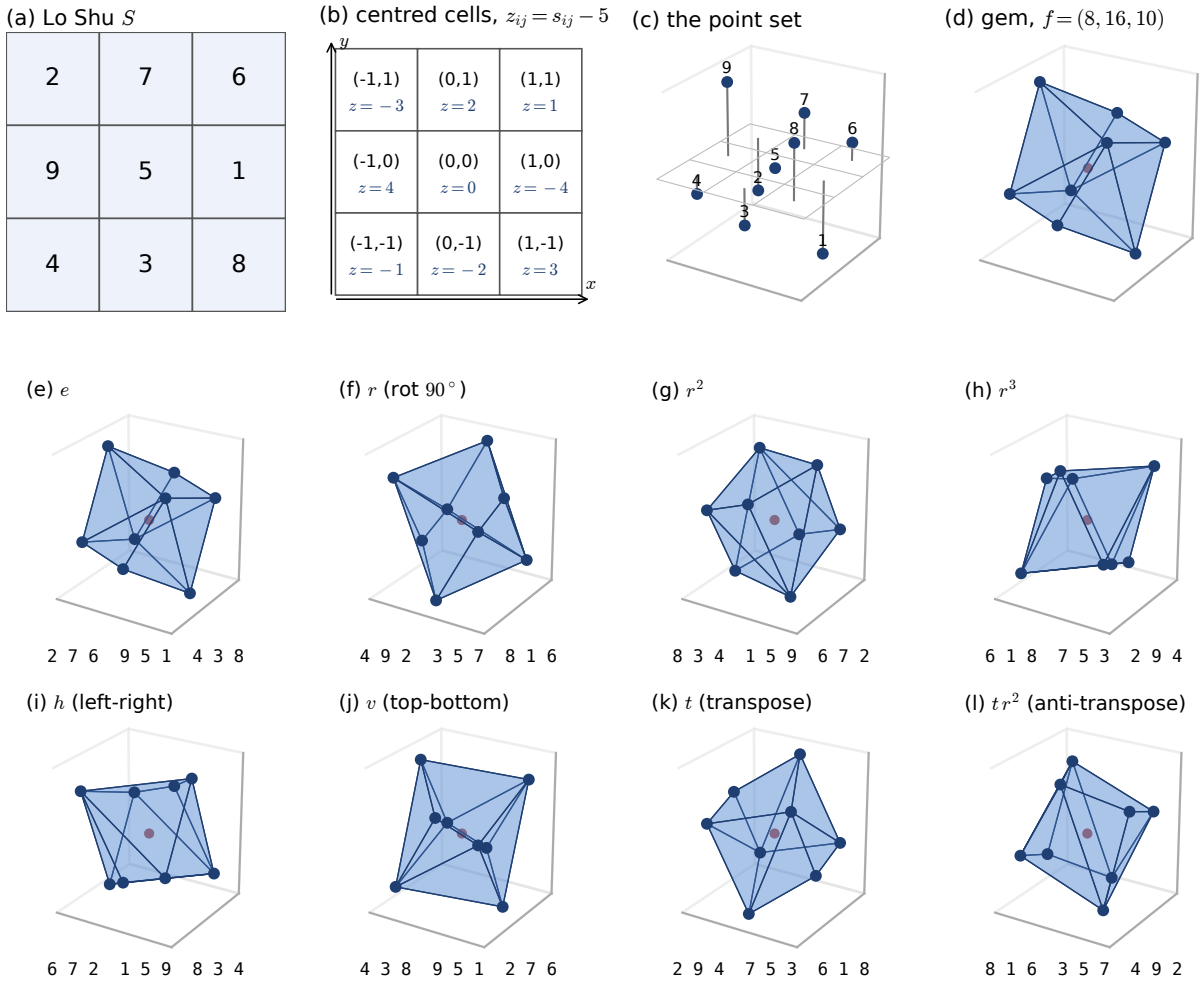


Figure 1: Construction of the Magic Gem and its D_4-invariance. Top row: (a) the Lo Shu square S; (b) the centred cells (x_{ij}, y_{ij}) with the heights $z_{ij} = s_{ij} - 5$; (c) the nine points $\mathbf{p}_{ij}$ over the grid, labelled by the entries s_{ij}; (d) their convex hull, the gem of Example 3.2, with f-vector $(8, 16, 10)$; the centre point $(0, 0, 0)$ is the only interior point. Rows two and three (e–l): the eight images of the Lo Shu under D_4—the identity e, the rotations r, r^2, r^3 by $90°, 180°, 270°$, the left–right and top–bottom reflections h, v, and the two diagonal reflections (transpose t and anti-transpose tr^2)—each printed row by row beneath its gem. All eight gems are drawn from the same viewpoint; by Proposition 3.3 they are the images of one another under the isometries $\hat{g}$, so they are congruent, and all have f-vector $(8, 16, 10)$ and volume 18.

4.1 Height functions, the inner product, and the energies

Let $\mathcal{F} = \mathbb{R}^{\text{cells}}$ be the N-dimensional space of real functions on the grid, with the normalised inner product and norm

$$\langle f, g\rangle = \frac{1}{N}\sum_{i,j} f_{ij} g_{ij}, \qquad \|f\|^2 = \langle f, f\rangle. \tag{5}$$

The mean of f is $\bar{f} = \langle f, \mathbf{1}\rangle$ and its centring is $\tilde{f} = f - \bar{f}\mathbf{1}$. For two functions ("test function" φ and height f) the empirical covariance over the N cells is $\mathrm{Cov}(\varphi, f) = \langle \tilde{\varphi}, \tilde{f}\rangle = \langle \tilde{\varphi}, f\rangle = \langle \varphi, \tilde{f}\rangle$; the last two equalities hold because $\tilde{\varphi} \perp \mathbf{1}$ and $\tilde{f} \perp \mathbf{1}$.

Lemma 4.1 (Means, variances, and covariance as an inner product)**.** *For every arrangement S of $\{1, \ldots, N\}$:*

1. $\bar{x} = \bar{y} = \bar{z} = 0$*, so z is centred and* $\mathrm{Cov}(\varphi, Z) = \langle \tilde{\varphi}, z\rangle = \langle \varphi, z\rangle$ *for every $\varphi \in \mathcal{F}$;*

2. $\mathrm{Var}(X) = \mathrm{Var}(Y) = \dfrac{n^2 - 1}{12}, \quad \mathrm{Var}(Z) = \dfrac{N^2 - 1}{12} = \dfrac{n^4 - 1}{12}, \quad \mathrm{Cov}(X, Y) = 0;$

3. for every line L with indicator $\mathbf{1}_L$, $\quad \mathrm{Cov}(\mathbf{1}_L, Z) = \dfrac{\delta_L(S)}{N}$.

Proof. (1) $\sum_{i,j} x_{ij} = n\sum_j (j - \frac{n-1}{2}) = 0$, likewise for y, and $\sum_{i,j} z_{ij} = \sum_{v=1}^{N} v - N\frac{N+1}{2} = 0$; then $\mathrm{Cov}(\varphi, Z) = \langle \varphi, z\rangle - \bar{\varphi}\langle \mathbf{1}, z\rangle = \langle \varphi, z\rangle$. (2) $\mathrm{Var}(X) = \frac{1}{n}\sum_j (j - \frac{n-1}{2})^2$ is the variance of the uniform distribution on n equally spaced points, $(n^2-1)/12$; the same for Y, and for Z with N points. $\sum_{i,j} x_{ij} y_{ij} = (\sum_j x_{\cdot j})(\sum_i y_{i\cdot}) = 0$. (3) By (1), $\mathrm{Cov}(\mathbf{1}_L, Z) = \frac{1}{N}\sum_{c\in L} z_c = \delta_L/N$ by (4). □

We write $R_k = \mathbf{1}_{\{i=k\}}$, $C_\ell = \mathbf{1}_{\{j=\ell\}}$, $D_{\mathrm{m}} = \mathbf{1}_{\{i=j\}}$, $D_{\mathrm{a}} = \mathbf{1}_{\{i+j=n-1\}}$ for the $2n+2$ line indicators.

Definition 4.2 (Complete and low-mode energies)**.** For an arrangement S with height function z,

$$E_{\mathrm{full}}(S) = \sum_{k=0}^{n-1} \mathrm{Cov}(R_k, Z)^2 + \sum_{\ell=0}^{n-1} \mathrm{Cov}(C_\ell, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2, \tag{6}$$

$$E_{\mathrm{low}}(S) = \mathrm{Cov}(X, Z)^2 + \mathrm{Cov}(Y, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2. \tag{7}$$

Note that (6) sums over *all* n rows and *all* n columns. The word "energy" is a label for these sums of squares and nothing more: no Hamiltonian is implied, and the temperature of the annealing in Section 6.2 is a search parameter only. By Lemma 4.1(3) it has the following elementary description, which we use constantly:

$$N^2 E_{\mathrm{full}}(S) = \sum_L \delta_L(S)^2 = \sum_k \rho_k^2 + \sum_\ell \gamma_\ell^2 + \delta_{\mathrm{m}}^2 + \delta_{\mathrm{a}}^2, \tag{8}$$

the sum of the squared line-sum deviations over all $2n+2$ lines. In particular $N^2 E_{\mathrm{full}}(S)$ is a non-negative integer.

Remark 4.3 (The four-term energy does not characterise magic squares for $n \geq 4$). $E_{\mathrm{low}} = 0$ imposes four linear conditions on the centred height functions, while "magic" imposes $2n$ independent ones (Lemma 4.5); for $n \geq 4$ the two cannot agree even before the permutation constraint is taken into account. A witness for $n = 4$ is

$$S = \begin{pmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{pmatrix},$$

whose diagonals both sum to 34 while the row sums are $(28, 38, 44, 26)$ and the column sums $(37, 25, 43, 31)$; nevertheless $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0$, so $E_{\mathrm{low}}(S) = 0$ although $E_{\mathrm{full}}(S) = 396/256$. Proposition 5.4 counts all such arrangements, and Remark 4.16 explains why none exist for $n = 3$.

The dihedral group D_4 acts on the grid by the rotations and reflections of the square, on arrangements by $(gS)_c = s_{g^{-1}c}$ and on functions by $(gf)_c = f_{g^{-1}c}$, so that $z(gS) = g\,z(S)$ and $\langle gf, gh\rangle = \langle f, h\rangle$. Every g permutes the $2n$ rows and columns among themselves and the two diagonals among themselves, and acts on the centred coordinates by a signed permutation: $\{gx, gy\} = \{\pm x, \pm y\}$ as unordered pairs up to sign.

Proposition 4.4 (D_4-invariance of the energies)**.** *For every $g \in D_4$ and every arrangement S: $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$ and $E_{\mathrm{low}}(gS) = E_{\mathrm{low}}(S)$.*

Proof. For any line L, $\delta_{gL}(gS) = \sum_{c\in gL} s_{g^{-1}c} = \delta_L(S)$, and $L \mapsto gL$ is a bijection of the set of lines, so the multiset of deviations is invariant and (8) gives $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$. For E_{low}, $\mathrm{Cov}(X, gZ) = \langle g^{-1}x, z\rangle$ with $g^{-1}x \in \{\pm x, \pm y\}$, and similarly for y, so $\mathrm{Cov}(X, gZ)^2 + \mathrm{Cov}(Y, gZ)^2$ is invariant; the diagonal terms are permuted among themselves. □

An earlier version of this paper omitted the last row and the last column from (6), on the grounds that $\sum_k \rho_k = \sum_\ell \gamma_\ell = 0$ makes them redundant. They are redundant for the zero set but not for the values: the reflection that maps column $n-1$ to column 0 changes that energy by $(\gamma_{n-1}^2 - \gamma_0^2)/N^2$, so it was not D_4-invariant. Keeping all $2n+2$ terms (two more than the dimension of the constraint space, Lemma 4.5) is the price of invariance; all E_{full} statistics in this paper use (6).

4.2 The projection lemma

Let $\tilde{R}_k = R_k - \frac{1}{n}\mathbf{1}$, $\tilde{C}_\ell = C_\ell - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{m}} = D_{\mathrm{m}} - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{a}} = D_{\mathrm{a}} - \frac{1}{n}\mathbf{1}$ be the centred line indicators (each line has n of the N cells, so each mean is $1/n$), and set

$$W_R = \operatorname{span}\{\tilde{R}_k\}, \quad W_C = \operatorname{span}\{\tilde{C}_\ell\}, \quad W_D = \operatorname{span}\{\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}\}, \quad W = W_R + W_C + W_D \subset \mathbf{1}^\perp.$$

Lemma 4.5 (Structure of W)**.** *For $n \geq 3$:*

1. *$W_R = \{f : \ f_{ij} = a_i, \ \sum_i a_i = 0\}$ and $W_C = \{f : \ f_{ij} = b_j, \ \sum_j b_j = 0\}$; both have dimension $n-1$.*
2. *$\langle \tilde{R}_k, \tilde{R}_{k'} \rangle = \frac{1}{N}(n\delta_{kk'} - 1)$, and the same for columns.*
3. *$W_R \perp W_C$, $W_R \perp W_D$, $W_C \perp W_D$.*
4. *$\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{m}} \rangle = \langle \tilde{D}_{\mathrm{a}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{n-1}{N}$ and $\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{\epsilon_n}{N}$ with $\epsilon_n = 0$ for n odd and $\epsilon_n = -1$ for n even; in particular $\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}$ are linearly independent and $\dim W_D = 2$.*
5. *$W = W_R \oplus W_C \oplus W_D$ is an orthogonal direct sum and $\dim W = 2n$.*

Proof. For two lines L, L', $N\langle \tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'} \rangle = |L \cap L'| - 1$, since each line has n of the $N = n^2$ cells. Two distinct rows are disjoint and a row meets itself in n cells (2); a row meets every column and each diagonal in exactly one cell, and so does a column (3); the two diagonals meet in the centre cell when n is odd and not at all when n is even (4), and the Gram matrix $\frac{1}{N}\left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ has positive determinant for $n \geq 3$. For (1), $\sum_k a_k \tilde{R}_k$ is the function $(i,j) \mapsto a_i - \bar{a}$, and every zero-mean function of i alone arises this way; the map $a \mapsto a - \bar{a}$ has rank $n-1$. (5) follows: the three subspaces are mutually orthogonal, hence independent, and $\dim W = 2(n-1) + 2$. □

For a subspace $V \subset \mathcal{F}$ let P_V denote the orthogonal projection with respect to (5).

Lemma 4.6 (Projection lemma)**.** *Let S be an arrangement with height function z and line deviations $\rho, \gamma \in \mathbb{R}^n$, $\delta = (\delta_{\mathrm{m}}, \delta_{\mathrm{a}}) \in \mathbb{R}^2$, and let $G = \left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ as in Lemma 4.5(4). Then*

1. *$P_{W_R} z$ is the function $(i,j) \mapsto \rho_i/n$ (the row means of z) and $P_{W_C} z$ is $(i,j) \mapsto \gamma_j/n$, with $\sum_k \operatorname{Cov}(R_k, Z)^2 = \|\rho\|^2/N^2 = \frac{1}{n}\|P_{W_R} z\|^2$ and likewise for columns;*
2. *$\|P_{W_D} z\|^2 = \frac{1}{N}\,\delta^{\mathsf{T}} G^{-1} \delta$, while $\operatorname{Cov}(D_{\mathrm{m}}, Z)^2 + \operatorname{Cov}(D_{\mathrm{a}}, Z)^2 = \delta^{\mathsf{T}}\delta/N^2$;*
3. *consequently*

$$E_{\text{full}}(S) = \frac{1}{n}\Big(\|P_{W_R} z\|^2 + \|P_{W_C} z\|^2\Big) + \frac{\delta^{\mathsf{T}}\delta}{N^2} = Q\big(P_W z\big), \tag{9}$$

where $Q(w) = \frac{1}{n}\|P_{W_R} w\|^2 + \frac{1}{n}\|P_{W_C} w\|^2 + \frac{1}{N^2} w_D^{\mathsf{T}} G^2 w_D$ (w_D the coordinates of $P_{W_D} w$ in the basis $\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}$) is a positive-definite quadratic form on W. In particular E_{full} depends on z only through $P_W z$; but $E_{\text{full}} \neq \|P_W z\|^2$ in general, because the centred indicators are not an orthonormal system.

Proof. (1) The orthogonal projection of z onto the functions of i alone is the function of row means $(i,j) \mapsto \rho_i/n$, which has zero mean because z does, so it is $P_{W_R} z$; its squared norm is $\|\rho\|^2/n^3$ while $\sum_k \mathrm{Cov}(R_k, Z)^2 = \|\rho\|^2/n^4$ by Lemma 4.1(3). Columns are identical. (2) The Gram matrix of $(\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}})$ is G/N and the vector of inner products with z is δ/N, so $\|P_{W_D} z\|^2 = (\delta/N)^{\mathsf{T}}(G/N)^{-1}(\delta/N) = \frac{1}{N}\delta^{\mathsf{T}} G^{-1}\delta$. (3) Sum (1) and (2). If $P_{W_D} z = w_{\mathrm{m}}\tilde{D}_{\mathrm{m}} + w_{\mathrm{a}}\tilde{D}_{\mathrm{a}}$ then $\delta = G w_D$, so $\delta^{\mathsf{T}}\delta/N^2 = w_D^{\mathsf{T}} G^2 w_D/N^2$; each piece of Q is positive definite on its orthogonal summand (G is invertible), so Q is positive definite on W. □

Lemma 4.7 (Zero set of the energies)**.** *For every arrangement S of $\{1,\dots,N\}$:*

1. $E_{\mathrm{full}}(S) = 0 \iff P_W z = 0 \iff z \perp W \iff$ *S is a magic square.*
2. $\sum_k \mathrm{Cov}(R_k, Z)^2 + \sum_\ell \mathrm{Cov}(C_\ell, Z)^2 = 0 \iff z \perp W_R \oplus W_C \iff$ *S is semi-magic.*
3. $E_{\mathrm{low}}(S) = 0 \implies$ *both diagonals sum to $M(n)$; the implication "$E_{\mathrm{low}}(S) = 0 \implies S$ magic" holds for $n = 3$ and fails for $n = 4$ (Remark 4.16, Proposition 5.4).*

Proof. (1) $E_{\mathrm{full}}(S) = 0$ iff $Q(P_W z) = 0$ iff $P_W z = 0$ iff $z \perp W$, and by (8) iff all line deviations vanish. (2) is the same with the diagonal terms removed; (3) is immediate from (7) and Remark 4.16. □

Remark 4.8 (What the projection lemma does and does not say). Lemma 4.7(1) is the definition of a magic square rewritten in the language of orthogonality; it is not a theorem about magic squares, and we do not present it as one. Its use is as a change of viewpoint, which makes three things available that are not visible in the line-sum formulation: (i) W can be replaced by polynomial test functions (Section 4.3), which produces a hierarchy of relaxations whose degree-two level implies Loly's inertia invariance; (ii) for a uniformly random arrangement the quantities $\langle \varphi, z\rangle$ are permutation statistics in the sense of Hoeffding, with computable moments and limit law (Section 5.2); (iii) a single swap changes z by a multiple of a difference of two cell indicators, and (8) turns the change of E_{full} into a two-line count (Section 4.4).

4.3 The moment hierarchy

The functions x and y are the degree-one polynomials in the column and row index. Replacing them by higher powers gives a natural family of relaxations of the semi-magic condition.

Definition 4.9 (Degree-k conditions)**.** Let $f \in \mathcal{F}$ be centred ($\langle f, \mathbf{1}\rangle = 0$) and $k \geq 1$. We say f satisfies the *degree-k conditions* if

$$\langle x^m, f\rangle = 0 \quad \text{and} \quad \langle y^m, f\rangle = 0 \qquad \text{for all } 1 \leq m \leq k.$$

For $f = z(S)$ these read $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$ for $1 \leq m \leq k$, by Lemma 4.1(1).

Write $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ for the n distinct values taken by x and y, and for centred f let $\gamma_j(f) = \sum_i f_{ij}$ and $\rho_i(f) = \sum_j f_{ij}$ be its column and row sums (for $f = z(S)$ these are the deviations γ, ρ of (4)).

Proposition 4.10 (Moment hierarchy)**.** *Let $n \geq 3$ and let $f \in \mathcal{F}$ be centred.*

(a) For every function φ that depends only on the column index, $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; in particular $\langle x^m, f\rangle = \frac{1}{N}\sum_j x_j^m \gamma_j(f)$ and $\langle y^m, f\rangle = \frac{1}{N}\sum_i y_i^m \rho_i(f)$ for all $m \geq 0$. Hence if $\gamma(f) = 0$ and $\rho(f) = 0$ (for $f = z(S)$: if S is semi-magic), then $\langle x^m, f\rangle = \langle y^m, f\rangle = 0$ for every $m \geq 0$.

(b) If $k \geq n-1$, the degree-k conditions imply $\gamma(f) = \rho(f) = 0$.

(c) If $1 \leq k \leq n-2$, there is a centred f satisfying the degree-k conditions with $\gamma(f) \neq 0$; explicitly, $f_{ij} = \frac{1}{n}(-1)^j \binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $f_{ij} = 0$ for $j > k+1$. This f violates the degree-$(k+1)$ condition.

Thus, for real-valued height functions, the degree-k conditions are equivalent to "equal row sums and equal column sums" exactly when $k \geq n-1$.

Parts (b) and (c) are the invertibility of the Vandermonde matrix of the nodes $x_0, \dots, x_{n-1}$, and we claim no novelty for them; the threshold $n-1$ is the benchmark against which Theorem 4.14 is measured. Part (a) is, in physical language, the vanishing of the axial multipole moments $\sum_{ij} s_{ij} x_j^m$ of a semi-magic charge distribution, as Rogers and Loly observed for the dipole and quadrupole moments of a magic cube [20]. Note that the hierarchy uses the pure powers x^m, y^m only: the mixed moments $\sum_{ij} s_{ij} x_j^a y_i^b$ with $a, b \geq 1$, beginning with the product of inertia $\sum_{ij} s_{ij} x_j y_i$, are not determined by the line sums (Remark 4.21).

Proof. (a) $\langle \varphi, f \rangle = \frac{1}{N} \sum_j \varphi(j) \sum_i f_{ij} = \frac{1}{N} \sum_j \varphi(j) \gamma_j(f)$; the row statement is symmetric. The last sentence is immediate.

(b) Suppose $\langle x^m, f \rangle = 0$ for $1 \leq m \leq k$ with $k \geq n-1$. Together with $\sum_j \gamma_j(f) = \sum_{i,j} f_{ij} = 0$ (the case $m = 0$), part (a) says that $\gamma(f) \in \mathbb{R}^n$ is orthogonal to the n vectors $(x_j^m)_j$, $m = 0, \dots, n-1$. These are the columns of the Vandermonde matrix of the n *distinct* nodes $x_0, \dots, x_{n-1}$, which is invertible; hence they span $\mathbb{R}^n$ and $\gamma(f) = 0$. The same argument with the nodes y_i gives $\rho(f) = 0$.

(c) Let $v_j = (-1)^j \binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $v_j = 0$ for $k+1 < j \leq n-1$ (this needs $k+1 \leq n-1$). For a polynomial p define the forward difference $(\Delta p)(t) = p(t+1) - p(t)$; Δ lowers the degree of a nonconstant polynomial by exactly one and kills constants, so $\Delta^{k+1} p = 0$ whenever $\deg p \leq k$. Writing E for the shift $(Ep)(t) = p(t+1)$ we have $\Delta = E - 1$, and since E and 1 commute the binomial theorem gives $\Delta^{k+1} = \sum_{j=0}^{k+1} \binom{k+1}{j} (-1)^{k+1-j} E^j$, i.e.

$$(\Delta^{k+1} p)(0) = (-1)^{k+1} \sum_{j=0}^{k+1} (-1)^j \binom{k+1}{j} p(j) = (-1)^{k+1} \sum_{j=0}^{n-1} v_j \, p(j). \tag{10}$$

Taking $p(t) = (t - \frac{n-1}{2})^m$ with $0 \leq m \leq k$ shows $\sum_j v_j x_j^m = 0$; in particular ($m = 0$) $\sum_j v_j = 0$. Now set $f_{ij} = v_j / n$. Then $\sum_{i,j} f_{ij} = \sum_j v_j = 0$, so f is centred; $\rho_i(f) = \frac{1}{n} \sum_j v_j = 0$ for every i, so all $\langle y^m, f \rangle$ vanish by (a); $\gamma_j(f) = v_j$, so $\langle x^m, f \rangle = \frac{1}{N} \sum_j v_j x_j^m = 0$ for $1 \leq m \leq k$ by (10); and $\gamma_0(f) = v_0 = 1 \neq 0$. Finally, for $p(t) = (t - \frac{n-1}{2})^{k+1}$ we have $\Delta^{k+1} p \equiv (k+1)! \neq 0$, so $\langle x^{k+1}, f \rangle = \pm (k+1)!/N \neq 0$. □

Corollary 4.11 (E_{low} is the degree-one truncation)**.** *$E_{\text{low}}(S) = \langle x, z \rangle^2 + \langle y, z \rangle^2 + (\delta_{\text{m}}^2 + \delta_{\text{a}}^2)/N^2$, so $E_{\text{low}}(S) = 0$ iff z satisfies the degree-1 conditions and both diagonals sum to $M(n)$ (Lemma 4.1). By Proposition 4.10(a) every magic square has $E_{\text{low}} = 0$; by Proposition 4.10(c) the degree-1 conditions characterise equal line sums among real height functions only when $n - 1 \leq 1$, i.e. never for $n \geq 3$.*

Corollary 4.12 (Loly's invariant from the hierarchy)**.** *Place a mass s_{ij} at the planar point (x_{ij}, y_{ij}) and let $I(S) = \sum_{i,j} s_{ij} (x_{ij}^2 + y_{ij}^2)$ be the moment of inertia about the centre [13]. Then for every arrangement*

$$I(S) = N \operatorname{Cov}\big(Z,\, X^2 + Y^2\big) + \frac{n^2(n^4-1)}{12} = \sum_j x_j^2 \gamma_j + \sum_i y_i^2 \rho_i + \frac{n^2(n^4-1)}{12}. \tag{11}$$

Hence $I(S)$ takes Loly's value $n^2(n^4-1)/12$ if and only if the single condition $\operatorname{Cov}(Z, X^2 + Y^2) = 0$ holds. This condition is implied by the degree-2 conditions of Definition 4.9, and therefore (Proposition 4.10(a)) by the semi-magic property, but it is much weaker than the degree-2 conditions (see below). The two halves $I_x = \sum_j x_j^2 (\gamma_j + M(n))$ and $I_y = \sum_i y_i^2 (\rho_i + M(n))$ each equal $n^2(n^4-1)/24$ for semi-magic squares.

The invariance and its validity for semi-magic squares are Loly's [13]: his derivation writes $I = I_x + I_y$ with $I_x = \sum_j x_j^2 \cdot (\text{column sum } j)$, $I_y = \sum_i y_i^2 \cdot (\text{row sum } i)$, uses the row and

column sums only, and gives $I_x = I_y = n^2(n^4-1)/24$. The corollary re-derives this; what it adds is the decomposition (11) into a degree-two cross-moment plus a constant, and the observation that the invariance is one scalar condition rather than the degree-two level. How much weaker the single condition is can be seen for $n = 3$ by enumeration of all 9! arrangements (`check_round1.py`): 24,192 of them have $I = 60$, against 72 satisfying the degree-2 conditions (the semi-magic squares, Remark 4.16). The values for $n = 3, \dots, 9$ are 60, 340, 1300, 3885, 9800, 21840, 44280 (OEIS A126275).

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$: $I(S) = \sum_{i,j} z_{ij}(x_{ij}^2 + y_{ij}^2) + \frac{N+1}{2}\sum_{i,j}(x_{ij}^2 + y_{ij}^2)$. The first sum is $N\mathrm{Cov}(X^2+Y^2, Z)$ by Lemma 4.1(1) and equals $\sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i$ by Proposition 4.10(a); the second is $\frac{N+1}{2}\cdot 2\cdot\frac{n^2(n^2-1)}{12} = \frac{n^2(n^4-1)}{12}$ by Lemma 4.1(2). For semi-magic squares $\gamma = \rho = 0$ and $I_x = M(n)\sum_j x_j^2 = \frac{n(n^2+1)}{2}\cdot\frac{n(n^2-1)}{12}$. □

Corollary 4.13 (Null distribution of Loly's invariant)**.** *For a uniformly random arrangement S of order n,*

$$\mathbb{E}[I(S)] = \frac{n^2(n^4-1)}{12}, \qquad \mathrm{Var}\big(I(S)\big) = N^2\sigma_n^2\,\mathrm{Var}(X^2+Y^2) = \frac{n^4(n^4-1)(n^2-4)}{1080}, \tag{12}$$

where $\sigma_n^2 = (n^2+1)/12$ and $\mathrm{Var}(X^2+Y^2) = (n^2-1)(n^2-4)/90$ is the variance of x^2+y^2 over the N cells. The mean equals Loly's value for every distribution of arrangements that is invariant under relabelling of the values $1,\dots,N$ (in particular the uniform distribution): magic and semi-magic squares sit exactly at the population mean of I, and the invariance says that their fluctuation vanishes, not that their value is extreme. For $n = 3$, $\mathrm{Var}(I) = 30$ (standard deviation 5.48; 29 values from 46 to 74); for $n = 4$, $\mathrm{Var}(I) = 2176/3$ (standard deviation 26.9), and 2.9% of arrangements have $I = 340$.

Proof. $N\mathrm{Cov}(Z, X^2+Y^2) = \sum_c \big(x_c^2+y_c^2-\overline{x^2+y^2}\big)z_{\pi(c)}$ is a linear permutation statistic with mean 0 (Section 2.3); by (2) with $a = x^2+y^2$ and $b = z$ its variance is $\frac{1}{N-1}\cdot N\mathrm{Var}(X^2+Y^2)\cdot N\mathrm{Var}(Z) = N^2\sigma_n^2\mathrm{Var}(X^2+Y^2)$, as in Proposition 5.1. Since X and Y are independent and uniform on n equally spaced points, $\mathrm{Var}(X^2+Y^2) = 2\mathrm{Var}(X^2) = 2(\mathbb{E}X^4 - (\mathbb{E}X^2)^2)$ with $\mathbb{E}X^2 = (n^2-1)/12$ and $\mathbb{E}X^4 = (n^2-1)(3n^2-7)/240$, which gives $(n^2-1)(n^2-4)/90$. The $n = 3$ values are confirmed by exhaustive enumeration and the $n = 4$ values by 2×10^7 Monte Carlo samples. □

Proposition 4.10 is a statement about real-valued height functions. For arrangements, i.e. for $f = z(S)$ with S a permutation of $\{1,\dots,N\}$, the integrality and boundedness of the line sums change the picture in an interesting way.

Theorem 4.14 (Integrality lowers the permutation threshold)**.** *For $n \geq 3$ let $k^*(n)$ be the least k such that every arrangement S of $\{1,\dots,N\}$ satisfying the degree-k conditions is semi-magic. Then $k^*(n) \leq n-2$ for all $n \geq 14$: for such n the degree-$(n-2)$ conditions already force equal row and column sums. By contrast $k^*(n) \leq n-1$ for all n (Proposition 4.10(b)), with equality for $n = 3, 4$: $k^*(3) = 2$ and $k^*(4) = 3$ by the witnesses*

$$n = 3,\ \textit{degree } 1: \quad \begin{pmatrix} 1 & 3 & 9 \\ 8 & 6 & 5 \\ 7 & 4 & 2 \end{pmatrix}, \qquad \gamma = (1,-2,1),\ \rho = (-2,4,-2);$$

$$n = 4,\ \textit{degree } 2: \quad \begin{pmatrix} 14 & 2 & 5 & 13 \\ 8 & 15 & 7 & 4 \\ 12 & 3 & 9 & 10 \\ 1 & 11 & 16 & 6 \end{pmatrix}, \qquad \gamma = (1,-3,3,-1),\ \rho = 0.$$

Proof. Lower bounds. For the 3×3 array, the column sums are $16, 13, 16$ and the row sums $13, 19, 13$, so $\gamma = (1,-2,1)$, $\rho = (-2,4,-2)$ and $\sum_j x_j\gamma_j = (-1)(1) + 0 + (1)(1) = 0$, $\sum_i y_i\rho_i =$

$(1)(-2)+0+(-1)(-2)=0$: the degree-1 conditions hold but the square is not semi-magic, so $k^*(3)\geq 2$. For the 4×4 array, every row sums to 34 and the column sums are $35, 31, 37, 33$, so $\rho=0$ and $\gamma=(1,-3,3,-1)$, which is the vector v of Proposition 4.10(c) with $k=2$; hence $\sum_j x_j^m\gamma_j=0$ for $m=1,2$ by (10), the degree-2 conditions hold, and the square is not semi-magic, so $k^*(4)\geq 3$.

The bound for $n\geq 14$. Let S satisfy the degree-$(n-2)$ conditions. By Proposition 4.10(a), γ is orthogonal to the $n-1$ linearly independent vectors $(x_j^m)_j$, $m=0,\dots,n-2$, so it lies in their one-dimensional orthogonal complement in $\mathbb{R}^n$. By (10) with $k+1=n-1$, this complement is spanned by $v_j=(-1)^j\binom{n-1}{j}$, $j=0,\dots,n-1$ (which is orthogonal to all polynomials of degree $\leq n-2$ and nonzero). So $\gamma=tv$ with $t=\gamma_0\in\mathbb{Z}$ because $v_0=1$ and γ is an integer vector. On the other hand every column sum of a permutation of $\{1,\dots,N\}$ lies between the sum of the n smallest and the sum of the n largest values, so

$$|\gamma_j|\leq \sum_{v=N-n+1}^{N} v-M(n)=\frac{n(2N-n+1)}{2}-\frac{n(N+1)}{2}=\frac{n(N-n)}{2}=\frac{n^2(n-1)}{2}.$$

If $t\neq 0$ then $\max_j|\gamma_j|\geq\binom{n-1}{\lfloor(n-1)/2\rfloor}$. We claim $\binom{n-1}{\lfloor(n-1)/2\rfloor}>\frac{n^2(n-1)}{2}$ for all $n\geq 14$. For $n=14$: $\binom{13}{6}=1716>1274=\frac{14^2\cdot 13}{2}$. Passing from n to $n+1$ multiplies the left side by $\binom{n}{\lfloor n/2\rfloor}/\binom{n-1}{\lfloor(n-1)/2\rfloor}$, which equals 2 for n even and $\frac{2n}{n+1}\geq\frac{28}{15}$ for odd $n\geq 14$, while it multiplies the right side by $\frac{(n+1)^2n}{n^2(n-1)}=\frac{(n+1)^2}{n(n-1)}\leq\frac{225}{182}<\frac{28}{15}$ for $n\geq 14$; the claim follows by induction. Hence $t=0$ and $\gamma=0$; the same argument applied to the rows gives $\rho=0$. □

Remark 4.15 ($k^*(n)=n-1$ for $5\leq n\leq 13$; computer-verified). For each $5\leq n\leq 13$ a randomised swap search found an arrangement with all row sums equal to $M(n)$ and column deviations $\gamma_j=(-1)^j\binom{n-1}{j}$; by (10) such an arrangement satisfies the degree-$(n-2)$ conditions without being semi-magic, so $k^*(n)=n-1$ for all $3\leq n\leq 13$. The $n=5$ witness is printed in Section C; all nine are in the ancillary file `revision_round2/appendix_data.json`, verified in exact integer arithmetic by `check_round1.py`. Thus the permutation threshold agrees with the linear-algebra threshold exactly for $n\leq 13$ and drops below it from $n=14$ on; determining $k^*(n)$ for $n\geq 14$ is open (Section 6).

Remark 4.16 (Why the low-mode energy appeared to work for $n=3$). Exhaustive enumeration of the 9! arrangements (`verify_theory.py`, exact arithmetic) gives 760 arrangements satisfying the degree-1 conditions, exactly 72 (the semi-magic squares) satisfying the degree-2 conditions, in accordance with $k^*(3)=2$, and exactly 8 satisfying the degree-1 conditions together with both diagonal conditions—the magic squares. So for $n=3$, $E_{\text{low}}(S)=0\iff S$ is magic, but for the "wrong" reason: the degree-1 conditions leave the families $\gamma=t(1,-2,1)$, $\rho=u(1,-2,1)$ open, and the two diagonal conditions combined with the permutation constraint happen to force $t=u=0$. Nothing of the sort survives for $n=4$: the arrangement of Remark 4.3 has $E_{\text{low}}=0$ with $\gamma=(3,-9,9,-3)$, $\rho=(-6,4,10,-8)$; it satisfies the degree-2 condition in x but not in y ($\sum_i y_i^2\rho_i=-28$).

4.4 The single-swap formula

Let $a\neq b$ be two cells and let S^{ab} be the arrangement obtained from S by exchanging the entries in a and b. Define

$$m(a,b)=\#\{\text{lines } L:\ L \text{ contains exactly one of } a,b\},$$
$$u_S(c)=\sum_{L\ni c}\delta_L(S)\qquad(\text{sum over the lines through } c).$$

Theorem 4.17 (Single-swap formula). *For every arrangement S and cells $a\neq b$, with $d=s_b-s_a$,*

$$N^2\big(E_{\text{full}}(S^{ab})-E_{\text{full}}(S)\big)=d^2\,m(a,b)+2d\,\big(u_S(a)-u_S(b)\big).\tag{13}$$

In particular, if S is a magic square then, with $d = |s_a - s_b|$,

$$E_{\text{full}}(S^{ab}) = \frac{d^2\, m(a,b)}{N^2}. \tag{14}$$

Proof. The swap changes the height function by $z(S^{ab}) = z(S) + d(\mathbf{1}_a - \mathbf{1}_b)$. For a line L, $\delta_L(S^{ab}) = \delta_L(S) + d\big(\mathbf{1}_L(a) - \mathbf{1}_L(b)\big)$: the deviation increases by d if L contains a but not b, decreases by d if L contains b but not a, and is unchanged if L contains both or neither. By (8),

$$\begin{aligned} N^2 \Delta E_{\text{full}} &= \sum_{L \ni a,\, L \not\ni b} \big[(\delta_L + d)^2 - \delta_L^2\big] + \sum_{L \ni b,\, L \not\ni a} \big[(\delta_L - d)^2 - \delta_L^2\big] \\ &= d^2 m(a,b) + 2d\Big(\sum_{L \ni a,\, L \not\ni b} \delta_L - \sum_{L \ni b,\, L \not\ni a} \delta_L \Big). \end{aligned}$$

Adding and subtracting the lines containing both a and b turns the bracket into $u_S(a) - u_S(b)$. If S is magic, all $\delta_L(S) = 0$, so $u_S \equiv 0$ and $E_{\text{full}}(S) = 0$, giving (14). □

Lemma 4.18 (The multiplicity $m(a,b)$)**.** *For distinct cells $a = (i,j)$, $b = (i',j')$:*

$$\begin{aligned} m(a,b) = 2 \cdot [i \neq i'] + 2 \cdot [j \neq j'] &+ [\textit{exactly one of } a,b \textit{ on the main diagonal}] \\ &+ [\textit{exactly one of } a,b \textit{ on the anti-diagonal}]. \end{aligned}$$

Hence $2 \le m(a,b) \le 6$, and $m(a,b) = 2$ if and only if a and b lie in a common row or a common column and neither of them lies on a diagonal. Such pairs exist for every $n \ge 3$ (for $n = 3$ exactly one pair up to D_4: the two non-central cells of the middle row or column; for $n \ge 4$ any two off-diagonal cells of a row).

Proof. If $i = i'$ the common row contains both cells and no other row contains either; if $i \neq i'$ the row of a contains a only and the row of b contains b only; columns likewise, and each diagonal contributes 1 or 0 according to whether it separates a from b. Since $a \neq b$ the row/column contribution is 2 or 4, so $2 \le m \le 6$. If $m = 2$ then $i = i'$ or $j = j'$ (not both) and no diagonal separates the cells; for $i = i'$, $j \neq j'$ both cells cannot lie on the same diagonal, so "not separated" means neither is on it. For existence: a row contains at most two diagonal cells, so it has at least $n - 2 \ge 2$ off-diagonal cells when $n \ge 4$, and the middle row of the 3×3 grid has the two off-diagonal cells $(1,0), (1,2)$. □

Corollary 4.19 (Strict local minimality and the exact gap)**.** *Let S be a magic square of order $n \ge 3$.*

1. *Every single swap strictly increases the energy: $E_{\text{full}}(S^{ab}) \ge 2/N^2 > 0$ for all $a \neq b$. Thus every magic square is a strict local minimum of E_{full} on the set of arrangements with respect to transpositions, for every n.*

2. *The smallest energy reachable by one swap is*

$$\Delta_{\min}(S) = \min_{a \neq b} \frac{(s_a - s_b)^2\, m(a,b)}{N^2}, \qquad \frac{2}{N^2} \le \Delta_{\min}(S) \le \frac{6}{N^2}.$$

   The lower bound is attained iff two cells with consecutive values share a row or column and are both off the diagonals; the upper bound is the swap of any two consecutive values ($d = 1$, $m \le 6$). Thus $\Delta_{\min}(S) = \Theta(n^{-4})$ with explicit constants, uniformly over all magic squares of order n.

3. *The possible values of $N^2 E_{\text{full}}(S^{ab})$ are the integers $d^2 m$ with $1 \le d \le N-1$ and $2 \le m \le 6$. For the 880 magic squares of order 4 the multiset of $N^2 \Delta_{\min}$ is $\{2 : 432,\ 3 : 420,\ 4 : 28\}$, and for the 8 squares of order 3 it is $\{3 : 8\}$ (Section 5.3).*

Proof. By (14), $N^2 E_{\text{full}}(S^{ab}) = d^2 m(a,b)$ with $d \geq 1$ and $m \geq 2$ (Lemma 4.18); the cells carrying the values $v, v+1$ have $d = 1$ and $m \leq 6$, and the equality case of the lower bound is Lemma 4.18. □

Remark 4.20 (Swap formula for E_{low}). Since $\text{Cov}(\varphi, Z^{ab}) = \text{Cov}(\varphi, Z) + \frac{d}{N}(\varphi(a) - \varphi(b))$ for every test function φ (Lemma 4.1(1)), the same computation gives, for $E_{\text{low}}(S) = 0$ and in particular for S magic,

$$E_{\text{low}}(S^{ab}) = \frac{d^2}{N^2}\Big[(x_a - x_b)^2 + (y_a - y_b)^2 + \big(D_{\text{m}}(a) - D_{\text{m}}(b)\big)^2 + \big(D_{\text{a}}(a) - D_{\text{a}}(b)\big)^2\Big],$$

i.e. d^2/N^2 times the squared planar distance between the cells plus the number of diagonals separating them. The bracket is at least 1, so magic squares are strict local minima of E_{low} as well, with $E_{\text{low}}(S^{ab}) \geq 1/N^2$.

Remark 4.21 (The inertia tensor of the gem). Place unit masses at the N points $\mathbf{p}_{ij}$. The centre of mass is the origin (Lemma 4.1(1)) and, by (1) and Lemma 4.1(2), the inertia tensor has $I_{xx} = I_{yy} = N(\text{Var}Y + \text{Var}Z) = n^2(n^2-1)(n^2+2)/12$, $I_{zz} = N(\text{Var}X + \text{Var}Y) = n^2(n^2-1)/6$, $I_{xy} = -N\text{Cov}(X,Y) = 0$, $I_{xz} = -N\text{Cov}(X,Z) = -\sum_j x_j\gamma_j$ and $I_{yz} = -\sum_i y_i\rho_i$ (Proposition 4.10(a)). The diagonal entries are the same for all $N!$ arrangements, and for every arrangement satisfying the degree-1 conditions the tensor is diagonal: the unit-mass gem is an oblate symmetric top whose inertia carries no information about the square. The informative planar object is Loly's value-weighted one: its diagonal entries $\sum s_{ij}y_{ij}^2$, $\sum s_{ij}x_{ij}^2$ are fixed by the row and column sums (Corollary 4.12), whereas the product of inertia $P_{xy} = -\sum_{ij} s_{ij}x_{ij}y_{ij}$ is a mixed moment, fixed by no line sum: D_4 changes at most its sign, and over the 880 essentially different squares of order 4 the invariant $|P_{xy}|$ takes 48 distinct values between 0 and 60, vanishing for 56 of them.

5 Computations

This section reports the computations behind the statements of Section 4 and the two enumerative results of the paper. All counts and all swap gaps were computed in exact integer or rational arithmetic; Monte Carlo results are reported with standard errors. Hardware, software and algorithms are described in Section A.

5.1 Exhaustive enumeration for $n = 3$

All $9! = 362{,}880$ arrangements of order 3 were enumerated in exact integer arithmetic. Exactly 8 arrangements have $E_{\text{full}} = 0$ (the Lo Shu orbit), and the same 8 are the zeros of E_{low} (Remark 4.16); 760 arrangements satisfy the degree-one conditions, 4,224 have both diagonals summing to 15, and 72 are semi-magic. Both energies have mean 40/27 (the coincidence is special to $n = 3$: $\text{Var}(X) = \frac{n^2-1}{12}$ equals $n\text{Var}(C_j) = \frac{n-1}{n}$ only there), with standard deviations 0.4266 (E_{full}) and 0.8370 (E_{low}), modes 138/81 and 90/81, maxima 180/81 and 360/81, and 177 and 346 distinct values, all multiples of 1/81. Under single swaps the symmetric E_{full} has 16 local minima, the 8 magic squares and 8 non-magic arrangements at $E_{\text{full}} = 2/81$ (two line deviations of ± 1); E_{low} is more rugged, with 344 local minima, 336 of them non-magic.

5.2 The energy of a random arrangement

Let S be a uniformly random arrangement, i.e. the values $1, \ldots, N$ are placed in the grid by a uniformly random permutation π. For a test function φ on the grid write $T_\varphi := \text{Cov}(\varphi, Z) = \sum_c a_c\, z_{\pi(c)}$ with $a_c = (\varphi_c - \bar{\varphi})/N$; this is a linear permutation statistic in the sense of Section 2.3, and its first two moments follow from (2). Throughout, $\text{Var}(\varphi)$ and $\text{Cov}(\varphi, \psi)$ denote population variances and covariances over the N cells, and

$$\sigma_n^2 := \frac{\text{Var}(Z)}{N-1} = \frac{N+1}{12} = \frac{n^2+1}{12}, \qquad \text{Var}(Z) = \frac{N^2-1}{12}.$$

Proposition 5.1 (Exact moments of the random energies)**.** *For a uniformly random arrangement and any test functions φ, ψ,*

$$\mathbb{E}[T_\varphi] = 0, \qquad \mathbb{E}[T_\varphi T_\psi] = \sigma_n^2 \operatorname{Cov}(\varphi, \psi). \tag{15}$$

Consequently, with E_{low} as in (7) *and the symmetric E_{full} (all n rows, all n columns, both diagonals; $2n+2$ terms),*

$$\mathbb{E}\operatorname{Cov}(X,Z)^2 = \mathbb{E}\operatorname{Cov}(Y,Z)^2 = \frac{n^4-1}{144}, \qquad \mathbb{E}\operatorname{Cov}(D,Z)^2 = \mathbb{E}\operatorname{Cov}(R_k,Z)^2 = \frac{(n-1)(n^2+1)}{12\,n^2}, \tag{16}$$

$$\mathbb{E}[E_{\mathrm{low}}] = \frac{(n^2+1)\,(n^4-n^2+12n-12)}{72\,n^2} = \frac{n^4}{72} + \frac{n}{6} - \frac{13}{72} + \frac{1}{6n} - \frac{1}{6n^2}, \tag{17}$$

$$\mathbb{E}[E_{\mathrm{full}}] = \frac{n^4-1}{6\,n^2} = \frac{n^2}{6} - \frac{1}{6n^2}. \tag{18}$$

The covariance matrix $\Sigma_n := \sigma_n^2 G$ of the vector of covariances is σ_n^2 times the Gram matrix G of the centred test functions. For E_{low}, $G = \operatorname{diag}\big(\frac{n^2-1}{12}, \frac{n^2-1}{12}, \frac{n-1}{n^2}, \frac{n-1}{n^2}\big)$ except that the two diagonal indicators have $\operatorname{Cov}(D_{\mathrm{m}}, D_{\mathrm{a}}) = -1/n^2$ when n is even (they are uncorrelated when n is odd). For E_{full}, G is block diagonal: the n row indicators have Gram block $\frac{1}{n}(I - \frac{1}{n}J)$, the columns likewise, every row–column, row–diagonal and column–diagonal covariance vanishes, and the diagonal block is as for E_{low}. Hence the eigenvalues of Σ_n are

$$E_{\mathrm{low}}:\ \sigma_n^2\Big\{\tfrac{n^2-1}{12}\,(\times 2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\}, \qquad E_{\mathrm{full}}:\ \sigma_n^2\Big\{\tfrac{1}{n}\,(\times\, 2n-2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\} \tag{19}$$

for odd n; for even n the last pair $\{\frac{n-1}{n^2}, \frac{n-1}{n^2}\}$ is replaced by $\{\frac{1}{n}, \frac{n-2}{n^2}\}$. Finally, the variances are explicit rational functions of n, given as (21) *in Section B, with $\operatorname{Var}(E_{\mathrm{low}}) = n^8/5184 + O(n^7)$ and $\operatorname{Var}(E_{\mathrm{full}}) = n^3/36 + O(n^2)$.*

Proof. Since $\sum_c a_c = 0$, $\mathbb{E}[T_\varphi] = \bar{z}\sum_c a_c = 0$. For two linear permutation statistics $T_a = \sum_i a_i b_{\pi(i)}$, $T_{a'} = \sum_i a'_i b_{\pi(i)}$ the classical identity (2) reads $\operatorname{Cov}(T_a, T_{a'}) = \frac{1}{N-1}\sum_i (a_i - \bar{a})(a'_i - \bar{a}')\sum_j (b_j - \bar{b})^2$. With $a = (\varphi - \bar{\varphi})/N$, $a' = (\psi - \bar{\psi})/N$, $b = z$ this is $\frac{1}{N-1} \cdot \frac{N\operatorname{Cov}(\varphi,\psi)}{N^2} \cdot N\operatorname{Var}(Z) = \sigma_n^2 \operatorname{Cov}(\varphi,\psi)$, which is (15). The entries of G are elementary: $\operatorname{Var}(X) = \operatorname{Var}(Y) = \frac{n^2-1}{12}$; a line indicator has $\operatorname{Var}(L) = \frac{1}{n}(1-\frac{1}{n})$; two lines meeting in m cells have $\operatorname{Cov}(L, L') = \frac{m}{N} - \frac{1}{n^2}$ (Lemma 4.5); and $\operatorname{Cov}(X, Y) = 0$, $\operatorname{Cov}(X, D) = \operatorname{Cov}(Y, D) = 0$ since a diagonal contains every column and every row coordinate exactly once. Taking traces of Σ_n gives (16)–(18): $\mathbb{E}[E_{\mathrm{low}}] = \sigma_n^2\big(2 \cdot \frac{n^2-1}{12} + 2 \cdot \frac{n-1}{n^2}\big)$ and $\mathbb{E}[E_{\mathrm{full}}] = \sigma_n^2(2n+2)\frac{n-1}{n^2}$. The eigenvalues (19) follow from the block structure, $\frac{1}{n}(I - \frac{1}{n}J)$ having eigenvalue $\frac{1}{n}$ with multiplicity $n-1$ and 0 once.

The variances are obtained by expanding $\mathbb{E}[E^2]$ over the equality patterns of cell quadruples and using the exact fourth moments of a uniform permutation (Section A); the result is a rational number for each n, and the closed forms (21) were obtained by exact rational interpolation on $n = 3, \ldots, 45$, verified on 5–11 further orders each and against the exhaustive $n = 3$ values $\operatorname{Var}(E_{\mathrm{low}}) = 1532/2187$, $\operatorname{Var}(E_{\mathrm{full}}) = 398/2187$. □

Corollary 5.2 (Limiting law)**.** *As $n \to \infty$, the standardised vector*

$$\big(T_X/\sqrt{\Sigma_{11}},\ T_Y/\sqrt{\Sigma_{22}},\ T_{D_{\mathrm{m}}}/\sqrt{\Sigma_{33}},\ T_{D_{\mathrm{a}}}/\sqrt{\Sigma_{44}}\big)$$

converges in distribution to a standard Gaussian vector in $\mathbb{R}^4$, and therefore

$$\frac{E_{\mathrm{low}}}{(n^4-1)/144} \xrightarrow{\ d\ } \chi^2_2, \qquad \textit{equivalently} \qquad \frac{E_{\mathrm{low}}}{\mathbb{E}[E_{\mathrm{low}}]} \xrightarrow{\ d\ } \operatorname{Exp}(1). \tag{20}$$

For finite n the Gaussian approximation gives $E \approx \sum_j w_j \chi^2_{1,j}$ with independent χ^2_1 variables and weights w_j the eigenvalues (19)*; its density is computed by Imhof's inversion formula [12] in Figure 2.*

Proof. Each fixed linear combination $\sum_k \alpha_k T_{\varphi_k}$ is again a linear permutation statistic $\sum_c a_c z_{\pi(c)}$. Hoeffding's theorem [11, Thm. 3] (Section 2.3) asserts asymptotic normality provided that, for every $r \geq 3$, $N^{-1}\sum_{i,j}|d_N(i,j)|^r$ is $O\big((N^{-1}\sum_{i,j} d_N(i,j)^2)^{r/2}\big)$, d_N the doubly centred array. In the product case $c_N(i,j) = a_i z_j$ the ratio of the two sides equals $N^{1-r/2}\kappa_r(a)\kappa_r(z)$ with κ_r the standardised r-th absolute moments. For z (uniform on $\{1,\dots,N\}$, centred) $\kappa_r(z) = O(1)$; for each of X, Y and the two diagonal indicators, $\kappa_r(a) = O(n^{r/2-1}) = O(N^{r/4-1/2})$ (the worst case is an indicator of n of the N cells, a Bernoulli$(1/n)$ variable), so the ratio is $O(N^{1/2-r/4}) \to 0$. The same bound holds for every fixed combination $a = \sum_k \alpha_k a_k$ of the four standardised test functions: $\|a\|_r \leq \sum_k |\alpha_k| \|a_k\|_r$ by Minkowski's inequality, while $\mathrm{Var}(\sum_k \alpha_k T_k) \geq (1-o(1))\sum_k \alpha_k^2$ because the only nonzero correlation among the T_k tends to zero. By the Cramér–Wold device the vector is asymptotically Gaussian with covariance Σ_n; for even n the only nonzero correlation is $\mathrm{corr}(T_{D_\mathrm{m}}, T_{D_\mathrm{a}}) = -1/(n-1) \to 0$, so the standardised limit is $N(0, I_4)$. In $E_{\mathrm{low}} = \Sigma_{11}(\tilde{T}_X^2 + \tilde{T}_Y^2) + \Sigma_{33}\tilde{T}_{D_m}^2 + \Sigma_{44}\tilde{T}_{D_a}^2$ the diagonal terms carry weight $\Sigma_{33}/\Sigma_{11} = 12(n-1)/(n^2(n^2-1)) \to 0$, giving (20); the second form uses $\mathbb{E}[E_{\mathrm{low}}]/((n^4-1)/72) \to 1$. □

Remark 5.3. For E_{full} the number of statistics grows with n, so the corollary does not apply verbatim; the Gaussian mixture $\sigma_n^2\big[\frac{1}{n}\chi^2_{2n-2} + \frac{n-1}{n^2}\chi^2_2\big]$ is only a heuristic at the orders we can simulate, and the ratio of the exact to the Gaussian-mixture variance is $0.24, 0.45, 0.56, 0.64, 0.69, 0.73$ for $n = 3,\dots,8$ (for E_{low}: $0.51, 0.76, 0.86, 0.90, 0.93, 0.95$). For $n = 3$, E_{low} takes 346 and E_{full} 177 distinct values, all multiples of $1/81$; the Kolmogorov distances 0.10 and 0.20 at $n = 3$ are inflated by this discreteness and are not a failure of the Gaussian approximation as such.

n	energy	$\mathbb{E}[E]$ exact	MC mean	$\sqrt{\mathrm{Var}E}$ exact	MC std	Gauss. std	$\lvert\delta_{\mathrm{mean}}\rvert$	KS
3	E_{low}	$40/27 = 1.4815$	1.4814	0.8370	0.8369	1.171	6×10^{-5}	0.103
3	E_{full}	$40/27 = 1.4815$	1.4815	0.4266	0.4266	0.869	8×10^{-6}	0.200
4	E_{low}	$391/96 = 4.0729$	4.0736	3.1284	3.1288	3.586	2×10^{-4}	0.037
4	E_{full}	$85/32 = 2.6562$	2.6564	0.9074	0.9073	1.349	7×10^{-5}	0.107
5	E_{low}	$234/25 = 9.3600$	9.3605	8.0483	8.0482	8.694	6×10^{-5}	0.019
5	E_{full}	$104/25 = 4.1600$	4.1600	1.3966	1.3966	1.867	9×10^{-6}	0.078
6	E_{low}	$2035/108 = 18.843$	18.843	17.122	17.122	18.007	4×10^{-5}	0.012
6	E_{full}	$1295/216 = 5.9954$	5.9957	1.9671	1.9671	2.459	6×10^{-5}	0.061
7	E_{low}	$5050/147 = 34.354$	34.349	32.174	32.167	33.349	1×10^{-4}	0.008
7	E_{full}	$400/49 = 8.1633$	8.1631	2.5697	2.5698	3.089	2×10^{-5}	0.050
8	E_{low}	$22295/384 = 58.060$	58.058	55.377	55.381	56.888	3×10^{-5}	0.007
8	E_{full}	$1365/128 = 10.664$	10.664	3.2352	3.2353	3.777	3×10^{-5}	0.042

Table 1: Exact moments of Proposition 5.1 against Monte Carlo (10^8 uniformly random arrangements per order; for $n = 3$ the exact values were also confirmed by enumerating all 9! arrangements). $|\delta_{\mathrm{mean}}|$ is the relative error of the MC mean; all MC means and variances agree with the exact values to within 5×10^{-4} relative error, consistent with the sampling standard errors. "Gauss. std" is the standard deviation of the Gaussian-limit mixture $\sum_j w_j\chi_1^2$ of Corollary 5.2, and KS is the Kolmogorov distance between the MC sample and that mixture (not the Exp(1) limit).

5.3 Swap gaps: exact verification

Theorem 4.17 gives, for a magic square S and the swap of two cells a, b with $d = |s_a - s_b|$, $E_{\mathrm{full}}(S^{ab}) = d^2 m(a,b)/N^2$ with $m(a,b) \in \{2,\dots,6\}$ the number of lines separating a from b. We verified the formula in exact rational arithmetic, with the symmetric energy evaluated directly from the definition, on every one of the $880 \cdot \binom{16}{2} = 105{,}600$ single swaps of the 880 essentially different order-4 squares, the $8 \cdot \binom{9}{2} = 288$ swaps of the eight order-3 squares, and all swaps of the Siamese squares of orders 5, 7 and 9: agreement in every case, and $E_{\mathrm{full}}(S) = 0$ exactly for every magic square. The gap spectrum is arithmetic, not "banded": over the 105,600 order-4 swaps $N^2\Delta E = d^2 m$ takes exactly 70 distinct integer values from 2 to $980 = 14^2 \cdot 5$, and the minimum gap $N^2\Delta_{\min}(S)$ is 2 for 432 squares, 3 for 420 and 4 for 28 (Corollary 4.19(3)); it is 3 for all eight order-3 squares and 3, 2, 2 for the Siamese squares of orders 5, 7, 9.

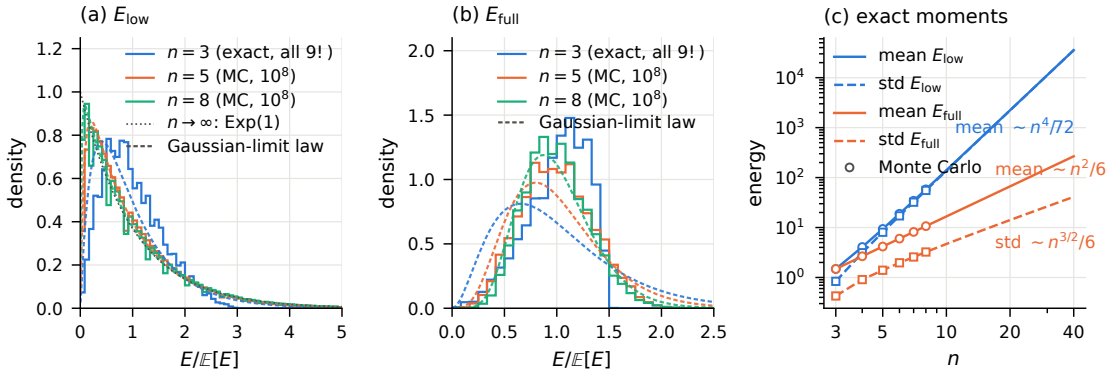


Figure 2: Energy of a uniformly random arrangement. (a) Distribution of $E_{\text{low}}/\mathbb{E}[E_{\text{low}}]$ for $n = 3$ (exact, all 9! arrangements) and $n = 5, 8$ (10^8 Monte Carlo samples each), with the Gaussian-limit mixture densities (dashed, Imhof inversion with the weights (19)) and the $n \to \infty$ limit Exp(1) of Corollary 5.2 (dotted). (b) The same for the symmetric E_{full}; here the Gaussian mixture is only a slowly improving approximation (see the remark after Corollary 5.2). (c) Exact mean (solid) and standard deviation (dashed) of both energies from Proposition 5.1, $3 \le n \le 40$, with the Monte Carlo values for $n = 3, \dots, 8$ (open markers). The mean of E_{low} grows like $n^4/72$ and its standard deviation like $n^4/72$ as well (an exponential limit has unit coefficient of variation); E_{full} has mean $\sim n^2/6$ and standard deviation $\sim n^{3/2}/6$.

5.4 The low-mode balanced class for $n = 4$

Write L_n for the set of arrangements of $\{1, \dots, N\}$ with $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = \mathrm{Cov}(D_{\mathrm{m}}, Z) = \mathrm{Cov}(D_{\mathrm{a}}, Z) = 0$, i.e. $E_{\text{low}} = 0$ (Corollary 4.11). Each of the four conditions is a *linear* statistic of the arrangement: with cells indexed by $(i, j) \in \{0, \dots, n-1\}^2$, $\mathrm{Cov}(X, Z) = 0 \iff T_1 := \sum_{i,j} j\, s_{ij} = \frac{n-1}{2} \cdot \frac{N(N+1)}{2}$, $\mathrm{Cov}(Y, Z) = 0 \iff T_2 := \sum_{i,j} i\, s_{ij}$ equals the same target, and the two diagonal conditions say that $T_3 = \sum_i s_{ii}$ and $T_4 = \sum_i s_{i,n-1-i}$ equal $M(n) = n(N+1)/2$.

Proposition 5.4 (Exact size of L_4)**.** $|L_3| = 8$ *(the Lo Shu orbit; the class with only* $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0$ *has* 760 *members), and*

$$|L_4| \;=\; 14\,639\,440 \;=\; 8 \times 1\,829\,930,$$

of which exactly 7040 *are magic squares. The class is* D_4*-invariant and* D_4 *acts freely on it (*1 829 930 *orbits of size* 8*). The first-moment class* $\{\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0\}$ *has* 6 697 177 424 *members for* $n = 4$*, and the class with both diagonals equal to* 34 *has* 49 932 288 000 *members.*

Computation. The count is a meet-in-the-middle enumeration over the four additive statistics $T_1, \dots, T_4$ (Section A; 9 s on 16 threads), with the matching pairs expanded explicitly to obtain the row- and column-sum breakdown of Table 2. Validation: on $n = 3$ the same program reproduces the brute-force counts 8, 760, 4224, 8 magic and 72 semi-magic; on $n = 4$, with the $2n$ line sums in place of T_1, T_2, it reproduces the 7040 magic and the 549,504 $= 8 \times 68{,}688$ semi-magic squares; the diagonal-only count was also obtained by direct combinatorics, and the probability $|L_4|/16!$ was checked by 2×10^9 Monte Carlo samples (1373 hits; relative error 2.7%). □

Table 2: Breakdown of L_4 by the number of row sums and column sums equal to $M(4) = 34$. Exactly two or three equal row sums are impossible: if $R_i = 34 + e_i$ then $\sum_i e_i = 0$ and $\sum_i i\, e_i = 0$ (this is T_2), which cannot hold with exactly one or two nonzero e_i; the same holds for columns. The table is symmetric under transposition.

	columns equal to 34			
rows equal to 34	0	1	4	total
0	9 637 312	2 172 824	100 728	11 910 864
1	2 172 824	429 312	9 336	2 611 472
4	100 728	9 336	**7 040** (magic)	117 104
total	11 910 864	2 611 472	117 104	14 639 440

Thus only $7040/14\,639\,440 = 0.048\%$ of the low-mode balanced arrangements are magic; 110 064 members have all row sums equal to 34 but not all column sums (and symmetrically), and 9 637 312 (66%) have *no* row or column summing to 34 at all; the arrangement of Remark 4.3 belongs to the last class.

Why random sampling never found a counterexample. $|L_4|/16! = 6.997 \times 10^{-7}$, about one arrangement in 1.43 million, so the estimate "below 10^{-8}" given in the previous version of this paper was too small by a factor of about 70. The 10^8-sample runs of Section 5.2 contain 61 members of L_4, against an expectation of 70.0 ± 8.4 (1.1 standard deviations low). The previous version reported that 6×10^7 samples for $n = 4$ contained no member of L_4, a 42-fold shortfall. On re-examination, the archived data file behind that statement holds 10^5 samples per order (the script's default), for which the expected number of members is 0.07, and we found no record of a larger run; the two other candidate causes are excluded (the energy evaluated by that code has the same zero set as E_{low}, and it evaluates to an exact floating-point zero on members of L_4, so no tolerance could have missed them). We attribute the earlier statement to a misreported sample size, but the cause cannot be reconstructed from the repository. A Gaussian local-limit estimate explains the order of magnitude of $|L_4|$: under a uniformly random arrangement the vector $(T_1, \ldots, T_4)$ has mean equal to the target vector (by symmetry) and covariance Σ given by (2) (T_1, T_2 are uncorrelated with each other and with the diagonals; the two diagonal sums have correlation $-1/(n-1)$ for even n and 0 for odd n), and the $\mathbb{Z}$-span of the change vectors of all single transpositions has index 1 in $\mathbb{Z}^4$ (Smith normal form, computed for $n \le 8$), so

$$|L_n| \;\approx\; N!\,(2\pi)^{-2}\det(\Sigma)^{-1/2}.$$

The estimate gives 13.6 for $n = 3$ (true: 8, ratio 0.59) and 1.82×10^7 for $n = 4$ (true: 14 639 440, ratio 0.80); for the two-condition classes the ratios are 0.59/0.91 (first moments) and 1.10/0.96 (diagonals), so the Gaussian approximation improves with n as the local limit theorem predicts. For $n = 5, 6$ we have no exact count; the estimate gives 6.7×10^{17} and 1.5×10^{33}, and Monte Carlo (10^{10} and 2×10^{10} samples) gives 5.1×10^{17} ($\pm 6\%$) and 1.2×10^{33} ($\pm 13\%$), conjectural orders of magnitude only. In every case the magic squares are a vanishing fraction of L_n: the four low-mode conditions cost a factor of order $\det(\Sigma)^{1/2} \propto n^{13}$, while a magic square requires $2n$ independent line conditions, each costing a further factor of order $n^{5/2}$.

5.5 Geometry of the 880 order-4 gems

For each of the 880 essentially different order-4 magic squares we computed the Magic Gem $\mathcal{G}(S)$ exactly (integer facet planes, merged coplanar triangles, rational volumes; Section A); Euler's relation holds for all 880 gems. Figure 3 and Table 3 summarise the results.

Vertex counts, f-vectors, volumes. Between 8 and all 16 of the points are hull vertices; $V = 12$ is modal (380/880). There are 39 distinct f-vectors, from $(8, 14, 8)$ (one gem, square #54)

Table 3: Face structure of the 880 order-4 gems by number of hull vertices V: number of squares, of distinct combinatorial types (face lattices up to isomorphism), of distinct exact volumes, and of distinct f-vectors (V, E, F). The full list of the 39 f-vectors with multiplicities is in the ancillary file `revision_round2/appendix_data.json`.

V	8	9	10	11	12	13	14	15	16	all
squares	7	10	103	80	380	106	162	14	18	880
types	4	1	27	32	129	52	85	7	15	352
volumes	4	5	51	33	111	40	65	6	14	155
f-vectors	4	1	6	5	7	5	5	3	3	39

and $(8, 16, 10)$ (one gem) to $(16, 36, 22)$ (nine gems), the modal one being $(12, 26, 16)$ (164 gems); the multiset of facet sizes (number of triangles, quadrilaterals, ...) is determined by the f-vector in every case. The exact volume takes 155 distinct values, from 58 (square #427) to 292/3 (square #379), mean 84.05. The denominators are $1, 2, 3, 6$: all three coordinates of every point are half-odd integers when n is even, so the translate of a gem by $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope of $\mathbb{Z}^3$ and its volume lies in $\frac{1}{6}\mathbb{Z}$. Surface areas range from 116.5 to 148.6.

Combinatorial types. Two gems have the same combinatorial type if their vertex–facet incidence structures (equivalently, their face lattices) are isomorphic. Grouping by a Weisfeiler–Lehman hash of the bipartite vertex–facet graph and confirming every group by an explicit graph isomorphism test gives

352 combinatorial types

among the 880 gems (the hash was already exact: no hash class had to be split). Their sizes are very uneven; the number of types with $1, 2, 3, \ldots$ members is

squares per type	1	2	3	4	5	6	7	8	10	13	14	16	26	32
types	93	201	8	20	4	7	4	5	5	1	1	1	1	1

(Figure 3c); the largest type, with $f = (10, 22, 14)$ (12 triangles, 2 quadrilaterals), is shared by 32 squares, and the second largest, $f = (12, 24, 14)$, by 26. The 7 gems with 8 vertices fall into 4 types, including a single $(8, 14, 8)$ gem with four triangular and four quadrilateral faces (square #54, Fig. 3d) and four $(8, 18, 12)$ gems sharing one simplicial type with 12 triangular faces. The 18 gems with all 16 points extreme belong to 15 types.

Congruence classes and injectivity. Two gems are *congruent* if an isometry of $\mathbb{R}^3$ (rotations, reflections, translations) maps one onto the other; since a distance-preserving bijection between finite point sets extends to an isometry, congruence is decided exactly by testing for a bijection of hull-vertex sets preserving all integer squared distances (Section A).

Proposition 5.5 (class → gem)**.** *1. The* 880 *hull-vertex sets are pairwise distinct: the map from D_4-classes of order-*4 *magic squares to hull-vertex sets (as subsets of $\mathbb{R}^3$) is injective. (For the full* 16*-point sets this is trivial, since the point set determines the arrangement.)*

2. Up to congruence there are exactly 616 *gems. The congruence classes are precisely the orbits of the complement map $S \mapsto 17 - S$, which acts on gems as the reflection $z \mapsto -z$:* 352 *squares are self-complementary (their gems are mirror-symmetric up to a lattice symmetry) and the other* 528 *form* 264 *mirror pairs. No two gems of non-complementary squares are congruent, not even as full* 16*-point sets.*

Part (2) is the only failure of injectivity, and it is the expected one: $17 - S$ is magic whenever S is, and its gem is the mirror image of $\mathcal{G}(S)$ in the plane $z = 0$. The 352 self-complementary classes are exactly Dudeney's group III (the 48 associated squares) together with all 304 squares of group VI [7]. The coincidence $352 = \#\{\text{types}\} = \#\{\text{self-complementary squares}\}$ is numerical only: the two partitions of the 880 are different.

Dudeney groups. Crossing the hull-vertex count with Dudeney's twelve groups (table in the ancillary files) shows that geometry is only loosely tied to Dudeney type: every group with more than eight members spans at least four vertex counts and at least 20 combinatorial types; the 8-vertex gems occur only in groups VI (three) and VIII (four).

Remark 5.6. Groups III and VI, the self-complementary ones, have only even vertex counts; this is forced: for a self-complementary square the reflection $z \mapsto -z$ composed with the element of D_4 that realises $17 - S$ is an isometry of the gem of order 2 or 4 without fixed hull vertices (no point has $z = 0$ when N is even), so hull vertices fall into orbits of size 2 or 4.

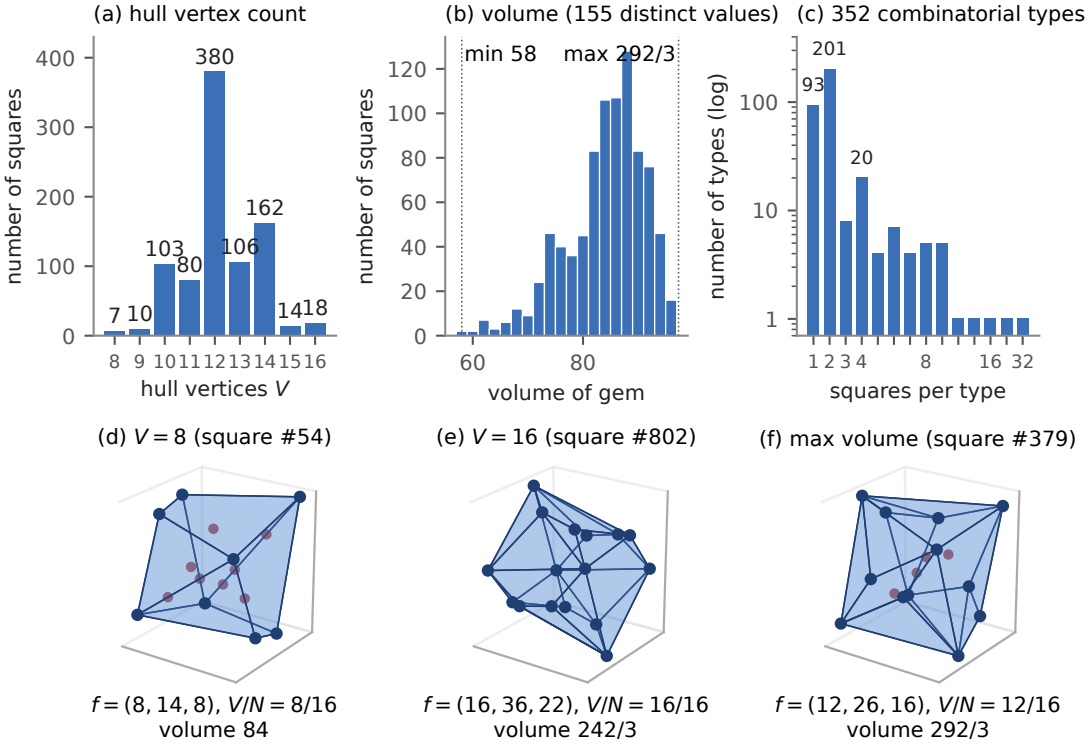


Figure 3: Geometry of the 880 order-4 Magic Gems. (a) Number of hull vertices (8 to 16; modal value 12). (b) Exact hull volume (155 distinct values between 58 and 292/3). (c) Sizes of the 352 combinatorial types (logarithmic scale): 93 types occur once, 201 twice, and one type is shared by 32 squares. (d)–(f) Three extreme gems, drawn with translucent faces after merging coplanar triangles: the unique $(8, 14, 8)$ gem (square #54; the eight interior points are shown in red), a gem with all 16 points extreme (square #802, Dürer's square with its first two rows exchanged and its last two rows exchanged), and the gem of largest volume (square #379).

Named squares. The gems of the Lo Shu, of Dürer's square and of the Siamese square of order 5 have hull fractions 8/9, 12/16 and 12/25 and exact volumes 18, 92 and 830/3. Dürer's own gem has twelve vertices, but exchanging its first two rows and its last two rows (which preserves the magic property) gives a square whose gem has all sixteen points extreme (square #802, Figure 3e): the hull-vertex count is a property of the square, not of the Dudeney group or of the construction method.

5.6 Siamese gems have twelve vertices

For the Siamese (de la Loubère) squares of odd order the hull-vertex count does not grow with n. We computed the gems for $n = 3, 5, \ldots, 21$: the Lo Shu gem has $f = (8, 16, 10)$, and every Siamese gem with $n \geq 5$ has $f = (12, 26, 16)$ with 12 triangular and 4 quadrilateral facets, so the hull fraction $12/n^2$ tends to zero. This is a theorem, with a short computer-assisted proof.

Proposition 5.7 (Siamese gems)**.** *Let $n = 2m+1 \geq 5$ and let S be the Siamese square of order n (1 in the middle cell of the top row, successive entries one step up and to the right modulo n, one step down when blocked). Then $\mathcal{G}(S)$ has exactly 12 vertices, 26 edges and 16 facets (12 triangles, 4 quadrilaterals). With 0-indexed cells (i, j) and $c = (n^2+1)/2$, the vertices are the points over the cells*

$$\begin{array}{lll} (0,0)\colon c+m+2, & (0,m-1)\colon n^2-1, & (0,m)\colon 1, \\ (0,2m)\colon c+m, & (m-1,0)\colon n^2-m, & (m,0)\colon m+2, \\ (m,2m)\colon n^2-m-1, & (m+1,2m)\colon m+1, & (2m,0)\colon c-m, \\ (2m,m)\colon n^2, & (2m,m+1)\colon 2, & (2m,2m)\colon c-m-2, \end{array}$$

i.e. the four corners, and on each side of the square the two cells holding the side's extreme values. For $n = 3$ the same construction gives the Lo Shu gem, with 8 vertices (all cells except the centre).

Proof. Step 1: closed form. Write $\langle t \rangle \in \{0, \ldots, n-1\}$ for t mod n. We claim $s(i,j) = n\langle i+j-m \rangle + \langle i+2j-2m \rangle + 1$. Indeed, along a run of n consecutive entries the move $(i,j) \to (i-1, j+1)$ fixes $i+j$ and increases $i+2j$ by 1 modulo n, while the "blocked" move $(i,j) \to (i+1,j)$ increases both $i+j$ and $i+2j$ by 1; the entry 1 sits at $(0,m)$ where both expressions vanish. Hence by induction the entry k lies in a cell with $\langle i+j-m \rangle = \lfloor (k-1)/n \rfloor$ and $\langle i+2j-2m \rangle = \langle k-1 \rangle$; since $(i,j) \mapsto (i+j,\, i+2j)$ is a bijection of $\mathbb{Z}_n^2$ (determinant 1), this determines the cell, which proves the claim.

Steps 2 and 3 (sketch; details in Section D). With $m = (n-1)/2$, $x = j-m$, $y = m-i$, the closed form gives $z = L(x,y) + n\,\zeta(x,y)$ with L affine and $\zeta = -n\lfloor (x-y+m)/n \rfloor - \lfloor (2x-y+m)/n \rfloor \in \{\pm(n+1), \pm 1, 0\}$. The affine bijection $\Phi(x,y,z) = (x, y, (z-L)/n)$ preserves the face lattice and the vertex cells, and $\Phi(\mathcal{G}(S))$ is the convex hull of the lattice points of five convex lattice polygons at the five heights ζ: two triangles T, $B = -T$ at $\zeta = \pm(n+1)$, two triangles U, $D = -U$ at $\zeta = \pm 1$, and a parallelogram M at $\zeta = 0$ (Figure 4a). Sixteen explicit planes, eight of them and their images under $(x,y,\zeta) \mapsto -(x,y,\zeta)$, are shown to support the convex hull of twelve named points $v_1, \ldots, v_{12}$—the three corners of T and of B, one corner each of U and D, and the four corners of M—in facets that close up into a surface with $(V,E,F) = (12, 26, 16)$ closed under edge-adjacency, hence all facets (so $P = Q$), and every vertex of every layer polygon satisfies all sixteen inequalities (Figure 4b). These are finitely many polynomial identities and inequalities in m, verified exactly for all integers $m \geq 2$ by the script `siamese_proof.py`. Hence $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{v_1, \ldots, v_{12}\}$, and mapping back by Φ^{-1} gives the statement; for $m = 1$ the four triangles degenerate to points and the same argument gives the 8-vertex Lo Shu gem. □

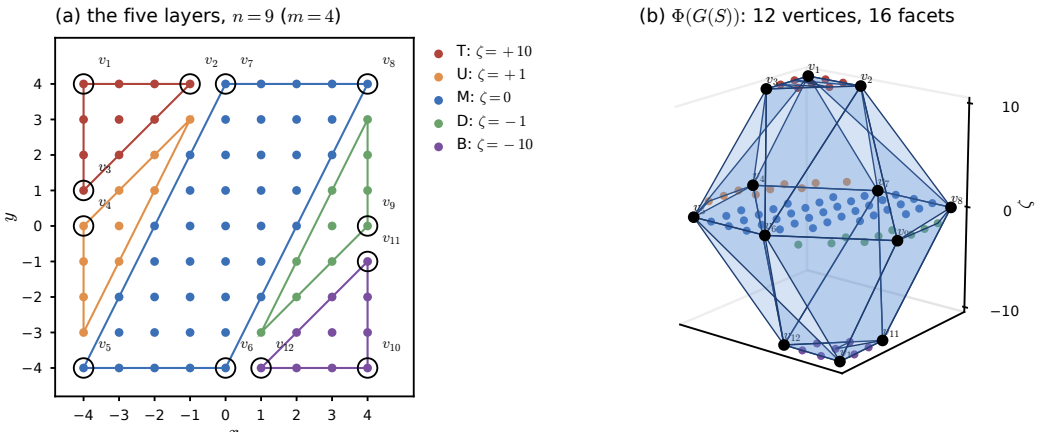


Figure 4: The Siamese gem of order $n = 9$ after the affine reduction of Proposition 5.7. (a) The 81 cells in the (x,y)-plane coloured by the layer value $\zeta \in \{+10, +1, 0, -1, -10\}$, with the five lattice polygons T, U, M, D, B outlined and the twelve vertex cells $v_1, \ldots, v_{12}$ circled. (b) The reduced polytope $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x,y,\zeta)\}$ with its 12 vertices and 16 facets (12 triangles, 4 quadrilaterals); the shapes of the layers do not depend on n except through scale, which is why the combinatorial type is constant for all odd $n \geq 5$.

6 Discussion

6.1 What the orthogonality viewpoint buys, and what it does not

Reading the line-sum conditions as $z \perp W$ changes nothing about which arrangements are magic, but it changes which questions are natural, and three of them have been answered above. First, the test space can be varied: replacing the $2n$ line indicators by the polynomials x^m, y^m $(m \le k)$ gives the moment hierarchy, Proposition 4.10 locates the degree $k = n-1$ at which polynomial test functions become equivalent to line sums for real heights, and Theorem 4.14 shows that the permutation constraint is not idle: for $n \ge 14$ integrality forces equal line sums one degree earlier than linear algebra does. This places two earlier observations in one frame: the four-term energy E_{low} is the degree-one member and is weaker than "magic" for every $n \ge 4$, while Loly's invariance of the moment of inertia [13] is a consequence of the degree-two member—the value $n^2(n^4-1)/12$ is the constant term of (11), the arrangement-dependent part is the single cross-moment $\operatorname{Cov}(Z, X^2+Y^2)$, and the invariance is weaker than the degree-two level (Corollary 4.12). Second, each $\operatorname{Cov}(\varphi, Z)$ is a linear permutation statistic, so the energies of a random arrangement have exact moments and a Hoeffding limit law (Proposition 5.1, Corollary 5.2), and Loly's moment of inertia has an exact null distribution (Corollary 4.13); the growth $\mathbb{E}[E_{\text{low}}] \sim n^4/72$, $\mathbb{E}[E_{\text{full}}] \sim n^2/6$ is a property of the normalisation $\operatorname{Cov} = \frac{1}{N}\sum$, not of the arrangements. Third, because $N^2 E_{\text{full}}$ is a sum of squared integer deviations, a swap changes it by the two-line formula (13), which gives local minimality and the exact gap spectrum for all n at once (Corollary 4.19).

What the viewpoint does not buy is equally clear. The projection lemma is a restatement of the definition (Remark 4.8); it proves nothing about magic squares that the line sums do not already say, and we have not used it to count anything: the counting is done by the meet-in-the-middle enumeration of Proposition 5.4 and by exact hull computations. The low-mode energy is a poor proxy for the magic property: for $n = 4$ its zero set is 2080 times larger than the set of magic squares, and even its two-condition sub-classes have 6.7×10^9 and 5.0×10^{10} members. The gem is injective on the 880 order-four classes only up to the complement map (Proposition 5.5), and its coarse invariants are not: the hull-vertex count takes nine values and the volume 155, and neither is well aligned with Dudeney's groups. Finally, the three-dimensional inertia tensor of the unit-mass gem is uninformative (Remark 4.21); the informative inertia statements are Loly's planar one and, for what varies across magic squares, the planar product of inertia.

6.2 Local search on the energy

By Theorem 4.17 a single swap changes E_{full} by an explicit quadratic expression in the swapped values, so local search is cheap. We ran two elementary searches on the symmetric E_{full} from uniformly random starting arrangements (Table 4): steepest swap descent (take the swap with the largest decrease; stop when none decreases the energy) and a plain simulated annealing (random swap proposals, Metropolis acceptance, geometric cooling from $T_0 =$ the median uphill increment of $N^2 E_{\text{full}}$ at a random arrangement to $T_1 = 1/2$, over $2 \times 10^4 N$ proposals). Descent reaches a magic square from 19.5% of random starts for $n = 4$ but from only 1.2% ($n = 5$) and 0.8% ($n = 6$); otherwise it stops after 8–15 swaps in a non-magic local minimum, typically with a single diagonal off by one (smallest residual $N^2 E_{\text{full}} = 1$ for $n = 5, 6$ and 2 for $n = 4$). Annealing succeeds in every run for $n = 4$ and in 78% ($n = 5$) and 50% ($n = 6$) of runs within the budget; its failures again end at $N^2 E_{\text{full}} \in \{1, 2\}$. The landscape is thus easy to descend but, as the exhaustive $n = 3$ count of non-magic local minima already suggests (eight, all at $E_{\text{full}} = 2/81$), it is riddled with diagonal-only defects that a swap move cannot repair. The experiment is small and the methods elementary; neither competes with the classical constructions or with the backtracking enumerations of [3, 22]. The energy is a search objective, not a counting tool.

	steepest swap descent (2×10^5 runs)			simulated annealing (400 runs)		
n	success	mean swaps	stuck ($E_{\text{full}} > 0$)	success	median proposals	accepted swaps
4	0.195 ± 0.001	8.1	80.5%	1.000	1.5×10^5	4.2×10^4
5	0.0115 ± 0.0002	11.2	98.9%	0.78 ± 0.02	4.2×10^5	8.0×10^4
6	0.0079 ± 0.0002	14.1	99.2%	0.50 ± 0.03	6.6×10^5	1.1×10^5

Table 4: Local search on the symmetric E_{full} from uniformly random arrangements. "Success" is the fraction of runs reaching $E_{\text{full}} = 0$ (a magic square), $\pm$ one standard error; "stuck" is the fraction of descents ending in a non-magic local minimum. Annealing budget $2 \times 10^4 N$ proposals; "median proposals" and "accepted swaps" refer to successful runs.

6.3 Open questions

1. *Size of the low-mode class.* $|L_3| = 8$, $|L_4| = 14{,}639{,}440$, and the Gaussian local-limit estimate $N!(2\pi)^{-2}\det(\Sigma)^{-1/2}$ is within 20% at $n = 4$ with ratio apparently increasing in n (Section 5.4). Is $|L_n|/N! \sim c\,n^{-13}$ with an explicit constant, i.e. does a local limit theorem hold for the four lattice statistics $(T_1, \dots, T_4)$?

2. *The permutation threshold.* What is $k^*(n)$ for $n \geq 14$? The proof of Theorem 4.14 compares the entries of the integer vectors orthogonal to all polynomials of degree $\leq k$ with the largest possible deviation $n^2(n-1)/2$. If the smallest such vectors were always alternating binomial vectors supported on $k+2$ consecutive columns, whose largest entry is $\binom{k+1}{\lfloor (k+1)/2 \rfloor} \approx 2^{k+1}/\sqrt{k}$, the comparison $2^k \gtrsim n^3$ would give $k^*(n) = O(\log n)$; but shorter integer vectors with small entries might exist, and we have no proof.

3. *Combinatorial types for $n = 5$.* The 34,413,153 order-five classes are within reach of the exact hull algorithm; the number of types, the modal f-vector and whether the complement map is again the only source of congruence are all computable.

4. *Bounded hull-vertex counts.* Siamese gems have twelve vertices for every odd $n \geq 5$ because, affinely, they are hulls of five lattice polygons whose shapes do not depend on n (Proposition 5.7). Do the LUX, Strachey or diagonal-complement constructions have the same property, and is the hull fraction of a uniformly random magic square of large order bounded away from zero?

5. *Other square families.* The construction and Section 4 apply verbatim to semi-magic, pandiagonal and Franklin squares [14], with W replaced by the span of the relevant line indicators; Loly's inertia invariance already covers the semi-magic case. The gems of these families have not been examined.

7 Conclusion

The Magic Gem is the convex hull of a magic square's height function over the centred grid, the same polyhedron for all eight D_4-images of the square. Reading the entries as a height function makes the magic conditions the orthogonality of that function to a $2n$-dimensional space of test functions; we have treated this as a viewpoint, not a result, and used it to obtain a moment hierarchy whose degree-two level implies Loly's inertia invariance and whose permutation threshold drops below the Vandermonde value $n-1$ from $n = 14$ on; exact moments and a limit law for the energies of a random arrangement, and the null distribution of Loly's moment of inertia; an exact single-swap formula with gap spectrum d^2m/n^4; the exact count 14,639,440 of 4×4 arrangements satisfying the four low-mode conditions; and an exact computation of the geometry of the 880 order-four gems, with a proof that Siamese gems have twelve vertices for every odd order. Several statements of the previous version were artifacts of an asymmetric energy and are withdrawn (Section 1). What remains is a small set of exact statements about a classical object, and a polyhedron that is pleasant to look at.

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A Computational methods

Hardware and software. All computations were run on an Apple M4 Max (16 cores, 128 GB) under macOS, with Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1 (Quickhull only seeds the exact hull computation) and C code compiled with Apple clang 17. Uniformly random arrangements were generated by Fisher–Yates shuffles (NumPy's default generator; `xoshiro256**` in C); the Monte Carlo of Table 1 uses 10^8 arrangements per order. All energies are evaluated from (6); the exact statements of Section 4 are checked by `verify_theory.py` in exact integer or rational arithmetic (all 105,888 single swaps of the $880 + 8$ squares, the $n = 3$ enumeration, the witnesses), and the checks added in this revision (Corollary 4.12, Corollary 4.13, the witnesses, the data of Section 5.4, the product of inertia) by `check_round1.py`.

Exact moments and variances. The means in Proposition 5.1 are traces of the covariance matrix $\sigma_n^2 G$. The variances were computed as exact rationals for each $n \le 45$ by expanding $\mathbb{E}[E^2]$ over the 15 equality patterns of a cell quadruple, with the fourth-moment expectations of the uniform permutation expressed through the power sums p_2, p_4 of the centred values; the rational functions (21) were then obtained by interpolation separately for odd and even n and verified on 5–11 further orders each, and against the exhaustive $n = 3$ values. Densities of the Gaussian-limit mixtures in Figure 2 were computed by Imhof's inversion formula [12].

Meet-in-the-middle enumeration of L_4. The four statistics $T_1 = \sum_{ij} j\, s_{ij}$, $T_2 = \sum_{ij} i\, s_{ij}$, $T_3 = \sum_i s_{ii}$, $T_4 = \sum_i s_{i,n-1-i}$ are additive over cells. The 16 cells are split into the top two rows A and the bottom two rows B. For each of the $\binom{16}{8} = 12{,}870$ value sets V and each of the 8! bijections $V \to A$ the partial vector $(T_1, \dots, T_4)|_A$ is packed into a 27-bit key; for the complementary set and each bijection $V^c \to B$ the key of (target − partial) is formed; the two key lists are sorted and merged, and the sum over matching keys of the product of multiplicities is $|L_4|$. This touches $2 \times 12{,}870 \times 40{,}320 \approx 1.04 \times 10^9$ partial vectors instead of $16! \approx 2.1 \times 10^{13}$ arrangements and runs in 9 s on 16 threads. Matching pairs are expanded explicitly to obtain the 14,639,440 members, which are then classified by row and column sums and by D_4-orbit. Validation is described after Proposition 5.4.

Exact hulls and their classification. Doubling all coordinates makes every point of a gem an integer vector. Facet planes are recovered from an initial Quickhull triangulation as primitive integer equations, coplanar triangles are merged into faces, vertices are the points on at least three facet planes, edges are vertex pairs sharing two facets, and volumes are exact rationals; Euler's relation $V - E + F = 2$ is checked for every gem. Combinatorial types are the isomorphism classes of the bipartite vertex–facet incidence graphs, grouped by a Weisfeiler–Lehman hash and confirmed by explicit isomorphism tests. Congruence is decided exactly by testing for a bijection of

hull-vertex sets preserving all integer squared distances (grouping by the sorted distance multiset, then an edge-labelled graph isomorphism), with a floating-point Kabsch superposition as a cross-check. For the Siamese proof (Proposition 5.7, Section D) the finitely many polynomial identities and inequalities in m are verified by exact integer polynomial arithmetic, an inequality $p(m) > 0$ for all integers $m \geq 2$ being certified by positivity of the coefficients of $p(2+t)$ or, failing that, by exact evaluation up to a Cauchy root bound.

Reproducibility. The scripts for every table and figure, the C enumerator and Monte Carlo code, the verification scripts, and the JSON files containing all reported numbers are provided as ancillary files, each directory with a summary of the numbers it produces and the commands that regenerate them. The web application at `https://kylemath.github.io/MagicGemWebpage/` displays the 880 essentially different order-four squares and renders the gem of a selected one.

B Variance closed forms

For a uniformly random arrangement the variances of the two energies of Proposition 5.1 are rational functions of n of the form

$$\begin{aligned} \mathrm{Var}(E_{\mathrm{low}}) &= \frac{(n-1)(n^2+1)\,P_{\mathrm{o}}(n)}{129600\,n^6}\ (n \text{ odd}), \qquad \frac{(n^2+1)\,P_{\mathrm{e}}(n)}{129600\,n^4}\ (n \text{ even}); \\ \mathrm{Var}(E_{\mathrm{full}}) &= \frac{(n-1)(n^2+1)\,Q_{\mathrm{o}}(n)}{180\,n^6}\ (n \text{ odd}), \qquad \frac{(n-2)(n^2+1)\,Q_{\mathrm{e}}(n)}{180\,n^5}\ (n \text{ even}), \end{aligned} \tag{21}$$

with integer polynomials P_{o}, P_{e} of degrees 11, 10 and leading coefficient 25, and Q_{o}, Q_{e} of degrees 6, 5 and leading coefficient 5, so that $\mathrm{Var}(E_{\mathrm{low}}) = n^8/5184 + O(n^7)$ and $\mathrm{Var}(E_{\mathrm{full}}) = n^3/36 + O(n^2)$. The polynomials are listed in the ancillary file `revision_round2/appendix_data.json` (and in `random_energy/variance_fits.json`); they were obtained as described in Section A, and at $n = 3$ they give 1532/2187 and 398/2187, the exhaustive values, at $n = 4$ 14093/1440 and 527/640 (Table 1).

C A witness for $k^*(5) = 4$

The arrangement

$$\begin{pmatrix} 1 & 22 & 11 & 24 & 7 \\ 23 & 3 & 12 & 2 & 25 \\ 18 & 6 & 19 & 17 & 5 \\ 4 & 21 & 14 & 10 & 16 \\ 20 & 9 & 15 & 8 & 13 \end{pmatrix}$$

has all row sums equal to $M(5) = 65$ and column sums $66, 61, 71, 61, 66$, i.e. $\gamma = (1, -4, 6, -4, 1) = \big((-1)^j \tbinom{4}{j}\big)_j$; by (10) it satisfies the degree-3 conditions of Definition 4.9 without being semi-magic, so $k^*(5) = 4$. The analogous witnesses for $6 \leq n \leq 13$ (Remark 4.15) are in the ancillary file `revision_round2/appendix_data.json`, together with the script that verifies them.

D Siamese gems: the remaining steps of the proof

We continue the proof of Proposition 5.7 from the closed form $s(i,j) = n\langle i+j-m\rangle + \langle i+2j-2m\rangle + 1$.

Step 2: affine reduction to five layers. In gem coordinates $x = j - m$, $y = m - i$, with $\langle t\rangle = t - n\lfloor t/n \rfloor$,

$$z = \underbrace{n(x-y+m) + (2x-y+m) + 1 - \tfrac{n^2+1}{2}}_{L(x,y)\ \text{affine}} + n\,\zeta(x,y), \quad \zeta := -n\left\lfloor \tfrac{x-y+m}{n} \right\rfloor - \left\lfloor \tfrac{2x-y+m}{n} \right\rfloor.$$

The map $\Phi(x,y,z) = (x, y, (z - L(x,y))/n)$ is an affine bijection, so $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x,y,\zeta)\}$ has the same face lattice and vertex cells as $\mathcal{G}(S)$. In the box $|x|, |y| \le m$ both floors lie in $\{-1, 0, 1\}$, the pairs $(-1,1), (-1,0)$ and, by the central symmetry $\zeta(-x,-y) = -\zeta(x,y)$, $(1,-1), (1,0)$ are impossible, so ζ takes the five values $\pm(n+1), \pm 1, 0$ on the lattice points of five convex lattice polygons (Figure 4a): $T = \mathrm{conv}\{(-m,m), (-1,m), (-m,1)\}$ at $\zeta = n{+}1$ (cut out by $y \ge x{+}m{+}1$), $U = \mathrm{conv}\{(-m, 1-m), (-m,0), (-1, m-1)\}$ at $\zeta = 1$ (by $2x + m + 1 \le y \le x + m$), $M = \mathrm{conv}\{(-m,-m), (0,m), (m,m), (0,-m)\}$ at $\zeta = 0$ (by $|y - 2x| \le m$), and $D = -U$, $B = -T$ at $\zeta = -1, -(n+1)$.

Step 3: the twelve points and sixteen planes. Let $v_1 = (-m, m, n{+}1)$, $v_2 = (-1, m, n{+}1)$, $v_3 = (-m, 1, n{+}1)$, $v_4 = (-m, 0, 1)$, $v_5 = (-m, -m, 0)$, $v_6 = (0, -m, 0)$, $v_7 = (0, m, 0)$, $v_8 = (m, m, 0)$, $v_9 = (m, 0, -1)$, $v_{10} = -v_1$, $v_{11} = -v_3$, $v_{12} = -v_2$ (the cells listed in the statement), and consider the inequalities

$$
\begin{aligned}
&(1)\ \zeta \le 2m + 2, \qquad (2)\ y \le m, \qquad (3)\ x \ge -m, \qquad (8)\ -2y + \zeta \le 2m,\\
&(4)\ -2(m{+}1)x + (2m{+}1)y - m\zeta \le m(2m{+}1),\\
&(5)\ 2(m{+}1)x - 2(m{+}1)y + (2m{+}1)\zeta \le 2m(m{+}1),\\
&(6)\ 2m(m{+}2)x - (2m^2{+}2m{-}1)y + m(2m{+}1)\zeta \le m(2m^2{+}2m{-}1),\\
&(7)\ 2mx - y + m\zeta \le m(2m{-}1),
\end{aligned}
$$

together with their images $(1')$–$(8')$ under $(x,y,\zeta) \mapsto -(x,y,\zeta)$; let Q be the polyhedron they define. The script `siamese_proof.py` verifies, as polynomial identities and inequalities in m valid for all integers $m \ge 2$ (Section A): (a) every vertex of T, U, M, D, B at its height satisfies all 16 inequalities, so $\Phi(\mathcal{G}(S)) \subseteq Q$; (b) each plane contains exactly the following v's, all others lying strictly inside:

$$
\begin{aligned}
&(1)\colon v_1v_2v_3, \quad (2)\colon v_1v_2v_8v_7, \quad (3)\colon v_1v_3v_5v_4, \quad (4)\colon v_1v_4v_7,\\
&(5)\colon v_2v_3v_6, \quad (6)\colon v_2v_6v_9, \quad (7)\colon v_2v_8v_9, \quad (8)\colon v_3v_5v_6,
\end{aligned}
$$

with centrally symmetric lists for $(1')$–$(8')$; (c) the quadrilaterals are in convex position in the cyclic order given, so the 16 face polygons have 26 distinct edges, each shared by exactly two polygons; (d) each v_k lies on three planes with independent normals. By (b) each plane supports $P := \mathrm{conv}\{v_1, \dots, v_{12}\}$ in a facet with the stated vertices; by (c) this family of facets is closed under edge-adjacency, and since the facet-adjacency graph of a 3-polytope is connected these are all the facets, so $P = Q$. With (a), $P \subseteq \Phi(\mathcal{G}(S)) \subseteq Q = P$, whose vertex set is $\{v_1, \dots, v_{12}\}$ by (d); mapping back by Φ^{-1} gives the statement, with $(V, E, F) = (12, 26, 16)$ read off from (b)–(c). For $m = 1$ the triangles degenerate to points and the same argument gives the 8-vertex Lo Shu gem.